

Office Hour Two: ...and Spaces

An annotated study edition

Original chapter: Matt Clay and Dan Margalit, *Office Hours with a Geometric Group Theorist* (Princeton University Press, 2017), pages 21–41.

How to read this

Each spread shows one original page on the left, untouched, and a column of annotations on the right. Orange numbered markers ① ② ③ on the page point to the line the annotation is about; a faint yellow bar marks the exact statement when it matters. Seven full-width visual pages (V1–V7) sit between the spreads and draw the constructions that the column is too narrow for. The level is the same as the chapter 1 edition: a strong high-school reader who has worked through Office Hour 1.

Colour code for the boxes

Big picture	where an idea sits in the whole book
Why?	the reason a definition or claim is true
Watch out	a common misreading, or a slip in the printed text
Try it	something to check by hand, and hints for exercises
Looking ahead	connections to later office hours
Background	facts from school mathematics the text assumes

Contents of the chapter

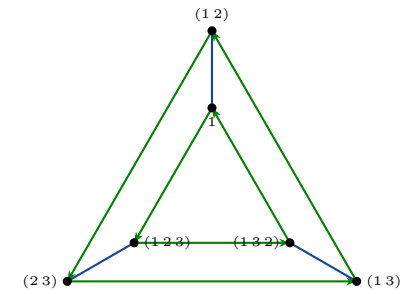
Group actions, orbits, Cayley’s theorem	book pp. 21–23	V1
2.1 Graphs, isomorphisms, automorphisms	book pp. 23–28	V2, V3
Cayley graphs and Theorem 2.2	book pp. 29–33	V4, V5
2.2 Metric spaces, word metric, isometries	book pp. 34–39	V6
Theorem 2.3 and 2.3 the big question	book pp. 40–41	V7
Exercise hints (all 26) and glossary	—	closing page

Big picture

The one sentence to keep in mind. Office Hour 1 said every group *should* be the symmetries of something; this office hour makes it true twice. First for graphs: G is exactly the symmetry group of its Cayley graph (Theorem 2.2). Then it turns the tables: with the word metric, G is *itself* a geometric object, and geometry can be used to prove algebra (Theorem 2.3).

Background you need

Assumed from Office Hour 1: group, subgroup, homomorphism, kernel, generating set, S_n , D_n , $\mathbb{Z}^n$, F_2 , reduced words. New school-level tools used here: absolute value and the triangle inequality $|a + b| \leq |a| + |b|$, Pythagoras, polar coordinates (r, θ) , the notion of a bijection between infinite sets.



S_3 with $\{(1\ 2), (1\ 2\ 3)\}$: green arrows = $(1\ 2\ 3)$,
blue = $(1\ 2)$ (two-way). A triangular prism.

A taste of what is coming: the Cayley graph of S_3 for the generators $(1\ 2)$ and $(1\ 2\ 3)$. Every group has such a picture, and the group is precisely its set of symmetries.

Office Hour Two

... and Spaces

Matt Clay and Dan Margalit



As we said at the outset of Office Hour 1, our goal is to exhibit every group as a collection of symmetries of some geometric object. We have given some examples of finite and infinite groups and in some cases we have given an appropriate geometric object with that group as its symmetries.

1 Our first task in this office hour is to say precisely what it means for a group to be a group of symmetries of a geometric object (answer: group actions). We will then define, for any group G , the Cayley graph of G and show that the group of symmetries of this graph is precisely the group G .

2 We will then introduce metric spaces, which formalize the notion of a geometric object. We'll see lots of metric spaces that groups can act on. Furthermore, we will see that groups themselves *are* metric spaces; in other words, groups themselves can be thought of as geometric objects. At the end we will use these ideas to frame the motivating questions of geometric group theory.

3 **Group actions.** In Office Hour 1 we got the idea that groups can *do* things to geometric objects (or sets) in the sense that elements of the group give symmetries of the object. For instance, each element of the dihedral group D_n does something to the n -gon, each element of the symmetric group S_n does something to the set $\{1, \dots, n\}$, and each element of $\mathbb{Z}$ does something to the real line. In other words, we can say that these groups *act* on those objects. This is precisely the idea of a group action.

1 The three moves of this office hour

(1) Make “ G is a group of symmetries of X ” precise: a *group action*. (2) Build, for any G , a graph whose symmetries are exactly G : the *Cayley graph*. (3) Introduce *metric spaces*, show that G itself is one, and state the questions that drive the whole book.

Big picture

Office Hour 1 went from groups to pictures example by example. Here the direction is reversed and made general: a single construction, valid for every group, produces the picture. That is why this chapter is more abstract than the first; the payoff is that from now on every group has a shape.

2 “Groups themselves are metric spaces”

The most important sentence on the page. A metric space is a set in which distances are defined (page 34). On page 37 the distance between two group elements g, h will be the length of the shortest word taking you from g to h . Everything from Office Hour 7 onwards depends on this idea.

3 “Does something to”

The examples are the ones you already know: a reflection in D_n moves the vertices of the polygon; a permutation in S_n moves the numbers $1, \dots, n$; an integer $n \in \mathbb{Z}$ slides the real line by n . In each case a group element is a *rule* that sends every point of the object to another point. “Acts on” is the word for this.

Background you need

A function of two variables $G \times X \rightarrow X$ takes a group element and a point and returns a point. Writing the output as $g \cdot x$ is just notation: it is not a product inside G (unless X happens to be G , as on page 23).

Try it

Before turning the page, write the two rules you would expect a “sensible” action to obey. (Hint: what should the identity do, and what should “do h , then do g ” equal?) Compare with the definition at the top of page 22.

Formally, an *action* of a group G on a set X is a function

$$1 \quad G \times X \rightarrow X$$

where the image of (g, x) is written $g \cdot x$ and where

- $1 \cdot x = x$ for all $x \in X$ and
- $g \cdot (h \cdot x) = (gh) \cdot x$ for all $g, h \in G$ and $x \in X$.

If A is a subset of X then $g \cdot A$ is the set $\{g \cdot a \mid a \in A\}$. When there is no confusion we sometimes write gx and gA instead of $g \cdot x$ and $g \cdot A$.

2 Notice that for any group G and any set X we always have the trivial action, defined by $g \cdot x = x$ for all $g \in G$ and $x \in X$.

Let S_X denote the symmetric group of X , that is, the group of permutations of X (this makes sense even if X is infinite). So not only does each element of G give rise to a permutation of X , but it does so in such a way as to respect the multiplications in the two groups.

3 The group S_X is the group of symmetries of X , thought of as a set (hence the name). An action of G on X is the same thing as a homomorphism $G \rightarrow S_X$. So an action of G on X is the formal way to realize G as a group of symmetries of the set X .

Exercise 1. Check that actions of G on X are the same as homomorphisms $G \rightarrow S_X$.

Here are some examples of group actions on sets:

- 4
1. S_n acts on $\{1, \dots, n\}$
 2. D_n acts on the set of vertices (or edges, or diagonals ...) of an n -gon
 3. $GL(n, \mathbb{R})$ acts on the set of vectors (or planes, or bases ...) in $\mathbb{R}^n$
 4. $\mathbb{Z}/2\mathbb{Z}$ acts on $\mathbb{Z}$ via multiplication by ± 1
 5. $\mathbb{Z}$ acts on $\mathbb{R}^n$ via $m \cdot v = A^m v$, where A is some fixed $n \times n$ matrix
 6. $\mathbb{Z}$ acts on the unit circle in $\mathbb{C}$ via the formula $m \cdot z = e^{im\theta} z$, where θ is any real number (so m rotates the circle by $m\theta$)

Exercise 2. List some other group actions.

5 If we have an action of a group G on a set X , then the *orbit* of $x \in X$ under G is the set

$$G \cdot x = \{g \cdot x \mid g \in G\}.$$

So, for example, the orbit of a vertex of an n -gon under the action of D_n on the set of vertices is the full set of all vertices; in such a situation we say that the action is *transitive*. At the other extreme, the orbit of x under the trivial action is simply $\{x\}$. If no nontrivial element of G fixes any $x \in X$, we say that the action of G on X is *free*.

Exercise 3. List all of the orbits for your favorite group actions (for instance, the ones listed above).

1 The two axioms

Axiom 1: the identity does nothing. Axiom 2: acting by h and then by g is the same as acting by the product gh . Note the order: $g \cdot (h \cdot x)$ applies h first, and gh is “ h then g ” in the convention of Office Hour 1, page 6. So axiom 2 says the action *respects the multiplication*.

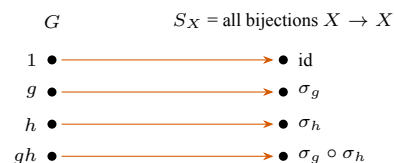
Watch out

Had we written $(hg) \cdot x$ in axiom 2 we would get a *right* action, a different notion. Right multiplication $g \cdot h = hg$ is *not* an action in the book’s sense (check: $g \cdot (h \cdot x) = xhg$ but $(gh) \cdot x = xgh$). The fix $g \cdot h = hg^{-1}$ works; try it.

2 The trivial action

Every element does nothing. It satisfies both axioms, which shows the axioms alone do not force anything interesting to happen; page 39 returns to this with the notion of the kernel of an action.

3 An action is a homomorphism $G \rightarrow S_X$



$\sigma_g(x) = g \cdot x$. Axiom 2 says $\sigma_g \circ \sigma_h = \sigma_{gh}$: the map $g \mapsto \sigma_g$ is a homomorphism. Axiom 1 says $\sigma_1 = \text{id}$, so $\sigma_{g^{-1}}$ undoes σ_g : each σ_g really is a bijection.

This is Exercise 1. The converse: given a homomorphism $\rho : G \rightarrow S_X$, set $g \cdot x = \rho(g)(x)$ and the axioms are exactly “ $\rho(1) = \text{id}$ ” and “ $\rho(gh) = \rho(g)\rho(h)$ ”. So the two descriptions are interchangeable, and one can pass freely between “ G acts on X ” and “ G maps to the symmetries of X ”.

4 The six examples

(4) ± 1 flips the sign. (5) $m \cdot v = A^m v$ needs A invertible, since m can be negative. (6) $e^{im\theta} z$ rotates z by the angle $m\theta$; if $\theta/2\pi$ is irrational no point ever returns, and the orbit of z is infinite and spread densely around the circle. Visual page V1 draws these.

5 Orbits, transitive, free

The orbit of x is everywhere x can be sent. *Transitive*: one orbit, everything reachable from everything. *Free*: no element except 1 has any fixed point.

Watch out

“Free” is a strong condition. D_n on the vertices of the n -gon is transitive but not free: a reflection whose axis passes through a vertex fixes that vertex. $\mathbb{Z}/2\mathbb{Z}$ on $\mathbb{Z}$ by ± 1 is not free because 0 is fixed. $\mathbb{Z}$ on the circle by an irrational rotation *is* free.

6 Exercise 3: hint

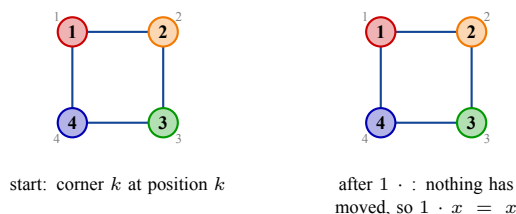
For each example list the orbits: S_n on $\{1, \dots, n\}$: one orbit. $\mathbb{Z}/2\mathbb{Z}$ on $\mathbb{Z}$: $\{0\}$ and the pairs $\{n, -n\}$. Rotations of the plane: circles centred at the origin, and the origin.

Interlude: A. Page 22: the two axioms of an action, and how to test a candidate rule

An action has three pieces of data: a group G , a set X , and a rule that takes a group element and a point of X and returns a point of X . The two axioms are promises about that rule. Pictures use the convention of Office Hour 1: grey numbers are fixed *positions*, and a coloured disc names the corner *originally* at that position, so a disc away from its grey number has moved. Here $G = D_4$ and X is the set of four corners of a square.

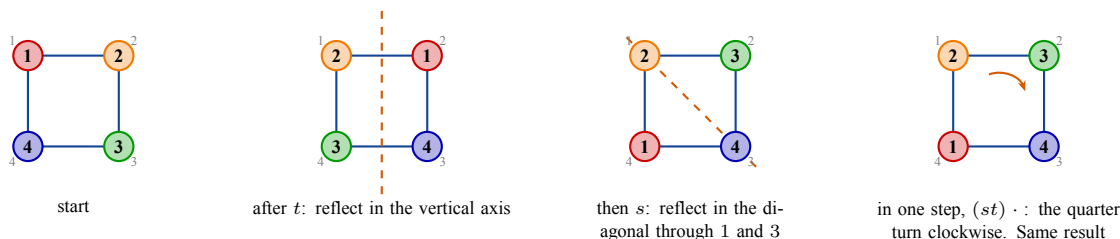
Axiom 1: the identity does nothing

$1 \cdot x = x$ for every x . This is not automatic: a rule can easily send everything to one fixed point, and such a rule is not an action.



Axiom 2: acting twice is acting by the product

$g \cdot (h \cdot x) = (gh) \cdot x$. Read the left side from the inside out: h moves the point first, then g moves the result. The right side does the single move gh , which by the convention of Office Hour 1 also means “ h , then g ”. Take $h = t$, the reflection in the vertical axis, and $g = s$, the reflection in the diagonal through positions 1 and 3; their product st is the quarter turn clockwise.



Why?

The second axiom is exactly what makes the assignment $g \mapsto (\text{the permutation } x \mapsto g \cdot x)$ a homomorphism $G \rightarrow S_X$, and the first says the identity goes to the identity permutation. So an action of G on X is a homomorphism $G \rightarrow S_X$, which is Exercise 1. Nothing is lost either way: given a homomorphism ρ , the rule $g \cdot x = \rho(g)(x)$ satisfies both axioms.

Big picture

Why bother with this definition at all? Office Hour 1 claimed that a group is a collection of symmetries and checked it example by example. Each check was improvised, because there was no definition of what “moves” means. The two axioms supply one, chosen so that $g \mapsto (x \mapsto g \cdot x)$ is a homomorphism $G \rightarrow S_X$. Four things follow at once.

- You can count.* Orbit–stabiliser turns group questions into counting questions, and you used it already without the name: the 24 cube rotations are 8 corners times 3 turns, that is $|\text{orbit}| \cdot |\text{stabiliser}| = |G|$.
- Office Hour 1 transfers wholesale.* An action is a homomorphism, so it has a kernel and an image, and no new machinery is needed.
- One group, many objects.* The group stops being identified with its symmetries, so one may ask which spaces it can act on.
- Geometry can force algebra.* Assume something about how a group acts, deduce algebra. Interlude M collects five theorems of that shape, including one whose statement mentions no

Testing a candidate rule

Most mistakes are caught by writing out both axioms once. In the table, G is any group and the rules are the ones people try.

X	rule $g \cdot x$	axiom 1	axiom 2
any	x (the trivial action)	✓	✓
G	gx (left multiplication)	✓	✓
G	xg	✓	✗
G	xg^{-1}	✓	✓
G	gxg^{-1} (conjugation)	✓	✓
$\mathbb{R}, G = \mathbb{Z}$	$x + g$	✓	✓
$\mathbb{R}, G = \mathbb{Z}$	gx	✗	✓
X	a fixed point x_0	✗	✓

Why?

Why $g \cdot x = xg$ fails and $g \cdot x = xg^{-1}$ works. For the first, $g \cdot (h \cdot x) = (xh)g = xhg$ while $(gh) \cdot x = xgh$, and in a non-abelian group these differ. For the second, $g \cdot (h \cdot x) = (xh^{-1})g^{-1} = xh^{-1}g^{-1} = x(gh)^{-1} = (gh) \cdot x$, so the inverse repairs the order. Multiplying on the right is naturally a *right* action; inverting turns it into a left one.

Why?

Why $g \cdot x = gx$ fails for $\mathbb{Z}$ acting on $\mathbb{R}$ by multiplication: the identity of $\mathbb{Z}$ is 0, so axiom 1 demands $0 \cdot x = x$, but the rule gives 0. The rule satisfies axiom 2, which shows the axioms are independent.

Try it

Check both axioms for D_n on the edges of the n -gon, and for S_n on the set of ordered pairs (i, j) with $i \neq j$ by $\sigma \cdot (i, j) = (\sigma(i), \sigma(j))$. Then decide whether $g \cdot x = g^{-1}x$ is an action.

The definition looks weak on purpose. The trivial action obeys both axioms and says nothing, so the axioms alone force nothing interesting. The work is done by the adjectives added afterwards: *transitive* and *free* in this office hour, *proper* and *cocompact* later. The payment is immediate, since the next twenty pages use the idea three times.

where	acts on	what comes out
p. 23	itself, by left multiplication	Cayley’s theorem: G sits inside a symmetric group
p. 31–33	its own Cayley graph	Theorem 2.2: those symmetries are exactly G
p. 40	$\mathbb{R}^n$, freely, by isometries	Theorem 2.3: G has no element of finite order

Interlude: L. What each idea in this office hour actually buys you

Each section of this office hour introduces a definition that looks like bookkeeping until you see what becomes possible with it. This page is the answer to “why are we doing this”, concept by concept, with the smallest convincing example in each case.

Group action

The gap it fills. Office Hour 1 kept saying that a group is a collection of symmetries, but never said what “acts on” means, so nothing could be proved in general.

What it buys. An action is a homomorphism $G \rightarrow S_X$, so kernel, image and injectivity all apply at once.

Smallest example. The 24 cube rotations acting on the four long diagonals give a homomorphism to S_4 ; it is injective and onto, and that single sentence is Office Hour 1’s main theorem restated.

Orbit and stabiliser

The gap it fills. Counting a group by hand is error-prone.

What it buys. $|\text{orbit}| \cdot |\text{stabiliser}| = |G|$, turning algebra into counting.

Smallest example. Count the cube’s rotations three ways and get the same answer each time:

acting on	orbit	stabiliser	product
8 corners	8	3	24
6 faces	6	4	24
12 edges	12	2	24

Lagrange’s theorem is the same statement: let G act on the cosets of H , and the orbit has size the index while the stabiliser is H .

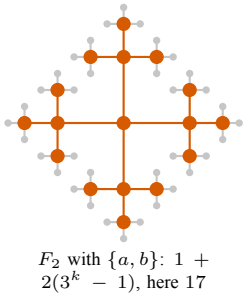
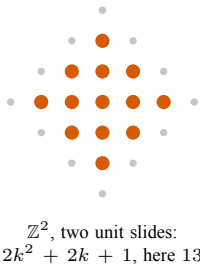
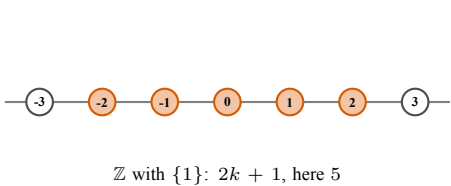
Graph

The gap it fills. A “geometric object” built from a group must be constructible from pure data, with no coordinates and no pictures assumed.

What it buys. Two sets, V and E , and already you have a shape, a notion of neighbour, and later a distance.

Smallest example. The Petersen graph is drawn three ways in the book and the drawings look unrelated; the data are identical.

The word metric in one picture: elements within distance 2 of the identity



Automorphism group

The gap it fills. “Symmetry” had meant a rigid motion, which needs distances. Graphs have none.

What it buys. Every graph now hands you a group, and *decorating* the graph shrinks that group in a controlled way.

Smallest example. The n -gon has $2n$ symmetries. Draw arrows on the edges, all one way round, and the reflections die: the count drops to n and the group is $\mathbb{Z}/n\mathbb{Z}$. That is how Office Hour 1 realised $\mathbb{Z}/n\mathbb{Z}$ geometrically.

Cayley graph

The gap it fills. Automorphism groups run the wrong way: a graph gives a group. The subject needs the reverse, a group giving a graph.

What it buys. A canonical object for every pair (G, S) , whose symmetries are exactly G (Theorem 2.2), and on which G therefore acts freely.

Smallest example. $\mathbb{Z}^2$ and F_2 both have two generators and nothing in the algebra makes the difference obvious. Their Cayley graphs are a flat grid and a tree, and the difference is visible instantly.

Metric space

The gap it fills. Graphs only have vertices and edges. Real geometric statements need distance, and “distance” should not mean “inside $\mathbb{R}^n$ ”.

What it buys. Distance, balls, shortest paths and an isometry group, on sets that are not made of numbers at all.

Smallest example. The distance between 0110101 and 1110000 is 3. Nothing here is geometric in the school sense, yet all three axioms hold and the space is the cube graph in disguise.

Word metric

The gap it fills. So far the group acted on a space from outside. The cornerstone claim of the office hour is stronger: the group *is* a space.

What it buys. Every group with a finite generating set becomes a metric space, so one may ask geometric questions about it: how fast do balls grow, what do geodesics look like, what does it look like from far away.

Smallest example. Count the elements within distance 2 of the identity in three groups. The three answers already separate the groups.

The strip at the foot of this page counts them.

Free and transitive

The gap it fills. The two action axioms permit the trivial action, so they force nothing.

What it buys. Adjectives strong enough to carry theorems. *Free* is the one that does the work in this office hour.

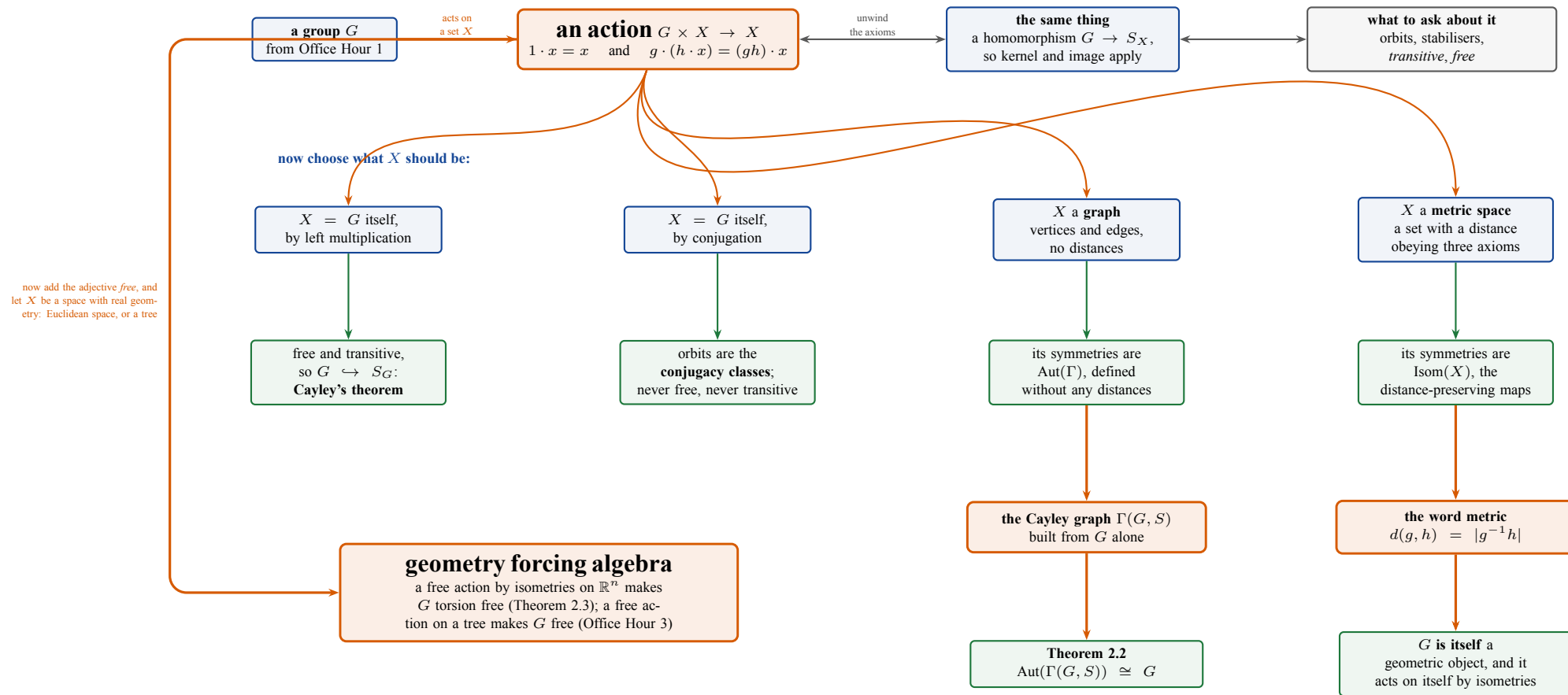
Smallest example. A group acting freely by isometries on Euclidean space has no element of finite order, which is Theorem 2.3. Drop “freely” and the statement collapses: the quarter turns of the plane form a group of order four acting by isometries.

Looking ahead

The facing page collects five theorems of exactly that shape, where a fact about an action produces a fact about the group.

Interlude: N. The map of this office hour: how the ideas fit together

Office Hour 2 has one central definition and four choices of what to feed it. Everything else on the map is a consequence of a choice. Read along the orange arrows: the action is the hub, the four boxes are the choices, and the green boxes are what each choice produces.



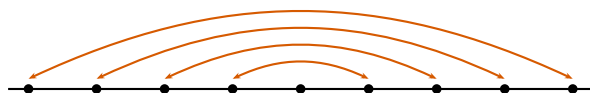
Why?

Why the two right-hand columns are the important ones. The first two choices of X take the group to act on itself, and they produce facts about groups: Cayley's theorem and the conjugacy classes. No geometry has appeared yet. The third and fourth choices introduce an object with shape, and they are what lets the last row happen: a graph built from the group alone, and a distance on the group itself. Office Hour 1 ended with a picture found by luck for each example; this chapter manufactures the picture.

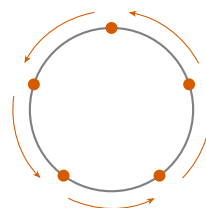
Big picture

The bottom left box is where the chapter is heading, and it is the reason for all the definitions above it. The pattern is always the same: assume a group acts on a space, assume one property of the action, and conclude something algebraic. Interlude M lists five instances, including one whose statement mentions no geometry whatever. The adjective that does the work in every case is *free*, which is why the map draws it entering the payoff box from the side.

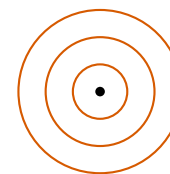
Interlude: V1. Actions and their orbits



$\mathbb{Z}/2\mathbb{Z}$ on $\mathbb{Z}$ by ± 1 : orbits $\{n, -n\}$;
0 is fixed, so the action is not free



$\mathbb{Z}$ on the circle, $\theta = 2\pi/5$: orbit of a point
has 5 points; with $\theta/2\pi$ irrational it is infinite



rotations of $\mathbb{R}^2$ about 0:
orbits are circles, and $\{0\}$

A group acting on itself (page 23). Two actions of G on the set $X = G$:

- *left multiplication* $g \cdot h = gh$. Transitive: from h you reach any k using $g = kh^{-1}$. Free: $gh = h$ forces $g = 1$. Every element moves every point, and the whole group is one orbit.
- *conjugation* $g \cdot h = ghg^{-1}$. The identity is fixed by everyone ($g1g^{-1} = 1$), so this action is never free (unless G is trivial) and never transitive (unless G is trivial). Its orbits are the *conjugacy classes*.

Why?

Why is conjugation an action? $g \cdot (h \cdot x) = g(hxh^{-1})g^{-1} = (gh)x(gh)^{-1} = (gh) \cdot x$, and $1 \cdot x = x$. Think of gxg^{-1} as “ x seen from g ’s point of view”: move by g^{-1} , do x , move back. In S_n , conjugating by g relabels the numbers: $g(1\ 2\ 3)g^{-1} = (g(1)\ g(2)\ g(3))$.

Conjugacy classes (Exercise 5). Conjugation in S_n relabels, so it preserves the cycle type; conversely any two permutations of the same cycle type are conjugate. For S_3 :

conjugacy class in S_3	elements	cycle type
identity	1	1 1 1
transpositions	(1 2), (1 3), (2 3)	2 1
3-cycles	(1 2 3), (1 3 2)	3

In D_n : r^k is conjugate to r^{-k} (conjugate by a reflection), and the reflections form one class if n is odd, two classes if n is even (axes through vertices versus axes through edge midpoints). In $\mathbb{Z}^n$, or any abelian group, $gxg^{-1} = x$, so every class is a single element. In F_n , two reduced words are conjugate exactly when, after trimming any $x \dots x^{-1}$ from the ends (“cyclic reduction”), one is a cyclic rotation of the other.

Looking ahead

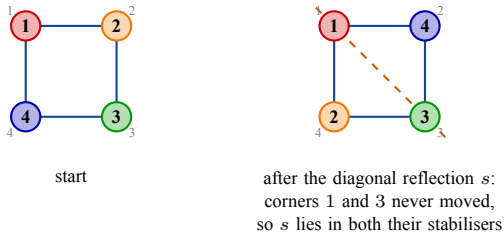
Conjugacy classes reappear as “how many essentially different symmetries are there”: in Office Hour 3 the conjugacy class of an element of F_2 tells you the shape of its action on the tree.

Interlude: B. Pages 22–23: orbits and stabilisers, and what “transitive” and “free” rule out

Three words describe an action, and all three are about *where points can go* and *what holds them still*. The **orbit** of x is $G \cdot x = \{g \cdot x : g \in G\}$, everywhere x can be sent. The **stabiliser** of x is $G_x = \{g \in G : g \cdot x = x\}$, everything that leaves x alone; it is a subgroup. The action is **transitive** if there is only one orbit, and **free** if no element except the identity fixes any point at all.

D_4 on the four corners: transitive, not free

Every corner can be carried to every other corner by a rotation, so there is one orbit and the action is transitive. But the reflection in the diagonal through positions 1 and 3 leaves those two corners exactly where they are, so that reflection is a non-identity element with fixed points: the action is not free.



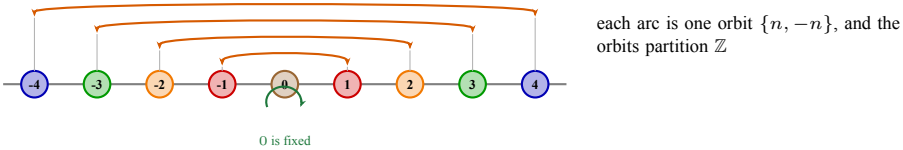
corner x	orbit $D_4 \cdot x$	stabiliser $(D_4)_x$	sizes
1	all four corners	$\{1, \text{diagonal reflection}\}$	$4 \cdot 2 = 8$
2	all four corners	$\{1, \text{other diagonal}\}$	$4 \cdot 2 = 8$

Why?

Orbit–stabiliser. $|\text{orbit}| \cdot |\text{stabiliser}| = |G|$ whenever G is finite, here $4 \cdot 2 = 8 = |D_4|$. The reason is the cosets of Office Hour 1: two elements send x to the same place exactly when they differ by an element of the stabiliser, so the orbit is in bijection with the cosets of G_x .

$\mathbb{Z}/2\mathbb{Z}$ on $\mathbb{Z}$ by ± 1 : neither transitive nor free

The non-identity element acts by $n \mapsto -n$. It fixes 0, so the action is not free; and no element carries 1 to 2, so there are many orbits and the action is not transitive.



The four combinations

G	X	rule	transitive	free
S_n	$\{1, \dots, n\}$	$\sigma \cdot i = \sigma(i)$	✓	✗ ($n \geq 3$)
D_n	corners of the n -gon	rigid motion	✓	✗
$\mathbb{Z}/2\mathbb{Z}$	$\mathbb{Z}$	± 1	✗	✗
G	G	gx	✓	✓
G	G	gxg^{-1}	✗	✗
$\mathbb{Z}$	$\mathbb{R}$	$x + g$	✗	✓
any G	any X	trivial	✗	✗

Watch out

“Free” is much stronger than “faithful”. An action is faithful when only the identity acts as the identity permutation; it is free when only the identity fixes even a single point. D_4 on the corners is faithful but not free.

Try it

List the orbits for: rotations of the plane acting on the plane; $\mathbb{Z}$ acting on the circle by irrational rotation; S_4 acting on the three ways of splitting $\{1, 2, 3, 4\}$ into two pairs. Which are transitive? Which are free? For D_4 on the four *edges*, compute a stabiliser and check orbit–stabiliser again.

Groups acting on themselves. If G is any group, then G acts on itself in two ways:

1. G acts on itself by left multiplication: $g \cdot h = gh$
 2. G acts on itself by conjugation: $g \cdot h = ghg^{-1}$
- The first action is transitive and free, but the second usually is not (the orbits in the second case are called *conjugacy classes*).

Exercise 4. Check that the purported actions of G on itself are really actions.

Exercise 5. Describe the conjugacy classes in S_n , D_n , F_n , and $\mathbb{Z}^n$.

The action of G on itself by left multiplication gives a homomorphism $G \rightarrow S_G$. This homomorphism is injective (why?). Therefore, we have the following theorem.

3. **THEOREM 2.1 (Cayley's theorem).** *Every group is isomorphic to a subgroup of some symmetric group.*

Cayley's theorem already tells us that every group can be thought of as a set of symmetries (of the group itself). However, there is no geometry here: we are just thinking of the group as a discrete set of points. The real goal is to find group actions on geometric objects so that we can use geometry to study the group. In this book the geometric objects we will think about are graphs and metric spaces, the subjects of the next two sections.

2.1 GRAPHS

We'll now describe our first class of geometric objects—graphs. We'll say what it means for a group to act on a graph, and finally we will show that every group is the group of symmetries of some graph, its Cayley graph.

4. **Graphs.** A *graph* Γ is a combinatorial object defined as follows: there is a nonempty set V (the set of vertices) and a set E (the set of edges), and a function from E to the set of unordered pairs of (not necessarily distinct) elements of V ; we'll call this function the *endpoint function*. We sometimes say that an edge *connects* its two endpoints.

We can draw a picture of Γ by drawing one dot for each element of V and connecting dots with arcs—one for each edge—according to the endpoint function; see, for instance, Figure 2.1.

Exercise 6. Write down a formal description of the graph in Figure 2.1, using sets V and E and the endpoint function.

5. The *valence* of a vertex in a graph is the number of times that vertex appears as the endpoint of an edge. In other words, this is the number of half-edges incident to the vertex. In Figure 2.1 every vertex has valence 4.

1 Left multiplication and conjugation

Both are verified on V1 (Exercise 4). Notice how different they are: one is as free as an action can be, the other has 1 as a fixed point of everybody. Left multiplication is the one that will give the Cayley graph its symmetries (page 30).

2 Conjugacy classes

Orbits of the conjugation action. See V1 for the answers to Exercise 5 in S_n , D_n , F_n and $\mathbb{Z}^n$.

3 Cayley's theorem

Why?

Why is $G \rightarrow S_G$ injective? Two elements $g \neq g'$ give different permutations because they disagree already at the point 1: $g \cdot 1 = g$ and $g' \cdot 1 = g'$. So the kernel is trivial and, by Office Hour 1 page 14, the homomorphism is injective; its image is a subgroup of S_G isomorphic to G .

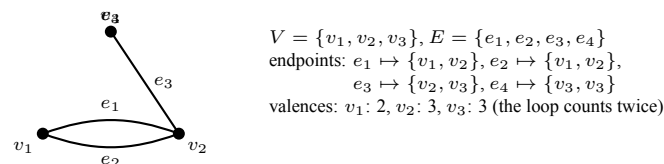
Example: $\mathbb{Z}/3\mathbb{Z}$ acting on itself by addition gives the three permutations $()$, $(0\ 1\ 2)$, $(0\ 2\ 1)$ of $\{0, 1, 2\}$: a copy of $\mathbb{Z}/3\mathbb{Z}$ inside S_3 .

Watch out

The symmetric group in the theorem is S_G , on $|G|$ points, which is enormous: D_4 (8 elements) lands inside S_8 , a group of 40,320 elements. The theorem is conceptually clean but geometrically empty, which is the point of the next paragraph: “there is no geometry here”.

4 Graphs, formally

A graph is two sets and a rule assigning to each edge its pair of endpoints. The definition deliberately allows an edge from a vertex to itself (a *loop*) and several edges between the same two vertices:



This is the kind of description Exercise 6 asks for.

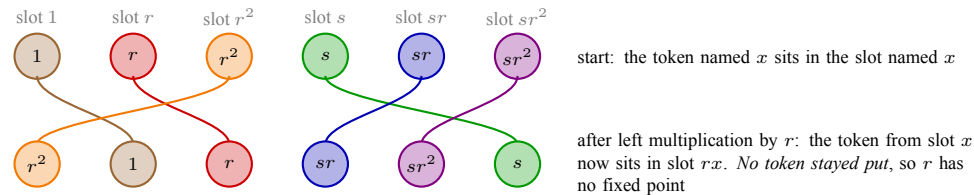
5 Valence

Count *half-edges* at the vertex: a loop contributes 2. In the example above, v_3 has valence 3 from one ordinary edge and one loop. Regular graphs (all valences equal) will be the important case: T_3 on page 25, T_4 on page 32.

Interlude: C. Page 23: the two ways a group acts on itself, and Cayley's theorem made concrete

Take $G = D_3 = S_3 = \{1, r, r^2, s, sr, sr^2\}$, with r a third of a turn and s a reflection. Draw six fixed *slots*, one per element, and put a token named x in the slot named x . An action of G on itself rearranges the tokens among the slots, and the picture of that rearrangement *is* the permutation of G that the acting element produces.

1. Left multiplication, $g \cdot x = gx$: transitive and free



Reading the second row as a permutation of the six slots gives $(1\ r\ r^2)(s\ sr^2\ sr)$: two three-cycles, no fixed point. That is the general picture.

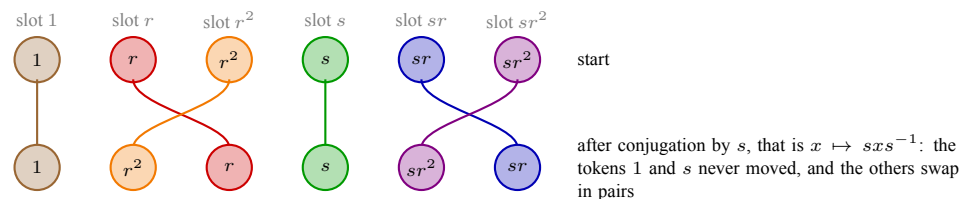
Why?

Transitive: given x and y , the element $g = yx^{-1}$ satisfies $g \cdot x = y$, so one orbit. *Free*: if $g \cdot x = x$ then $gx = x$, and cancelling x on the right gives $g = 1$. So the only element with any fixed point is the identity, which is exactly what makes this action the useful one.

Big picture

Cayley's theorem is this picture. Left multiplication assigns to each g a permutation of the set G , and axioms 1 and 2 say that the assignment $G \rightarrow S_G$ is a homomorphism. It is injective because g is recovered from its permutation by looking at where the token 1 went: $g \cdot 1 = g$. An injective homomorphism identifies G with its image, so *every group is a subgroup of a symmetric group*. For D_3 this realises a 6-element group inside S_6 . What the theorem does *not* give is geometry: the six slots are just six points in a row, with no shape. Supplying the shape is what the Cayley graph does on page 29.

2. Conjugation, $g \cdot x = gxg^{-1}$: usually neither



Conjugation by s fixes 1 and fixes s itself, so the action is not free; and since 1 can never be moved anywhere, it is not transitive either. The orbits of the conjugation action are the *conjugacy classes*, and they partition the group.

group	conjugacy classes
$S_3 = D_3$	$\{1\}, \{r, r^2\}, \{s, sr, sr^2\}$
S_n	one class per cycle type
D_n, n odd	$\{1\}$, the pairs $\{r^k, r^{-k}\}$, all reflections together
D_n, n even	as above, but the reflections fall into two classes
$\mathbb{Z}^n$	every element alone: $gxg^{-1} = x$
F_n	one class per cyclic rearrangement of a reduced word

Why?

Why the classes of S_n are the cycle types. Conjugating a permutation by τ relabels the numbers it acts on: $\tau\sigma\tau^{-1}$ has the same loop picture as σ with every entry replaced by its image under τ . Loop lengths therefore survive, and any relabelling can be achieved, so two permutations are conjugate exactly when their loops have the same lengths. In S_3 that gives the three classes above: the identity, the two three-cycles, the three swaps.

Watch out

Conjugation is an action on the *set* G , not a homomorphism on it. It fixes the identity always, so it can never be transitive for a group with more than one element, and it is free only for the trivial group. Its usefulness is elsewhere: it measures how far the group is from abelian, since the classes are single points exactly when G is abelian.

Try it

Draw the second row for left multiplication by s instead of r , and read off the permutation; it should be a product of three swaps. Then draw conjugation by r and check that it carries s to sr , so that the three reflections really do form one class. Finally, verify Cayley's theorem for $\mathbb{Z}/4\mathbb{Z}$: four slots, and the permutation given by adding 1.

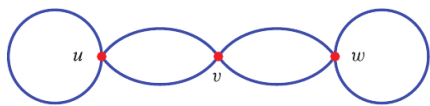


Figure 2.1 A graph.

1

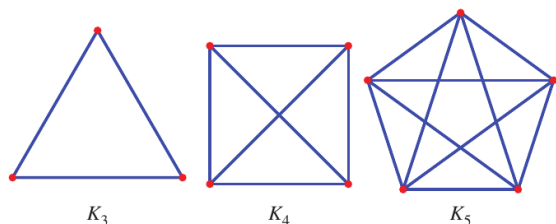
Graphs usually are described not by explicitly listing the sets V and E , but rather by giving some rule for listing the vertices and edges. Here are some examples:

2

1. The *complete graph* on n -vertices K_n has vertex set $\{1, \dots, n\}$ and an edge for every pair of distinct vertices.

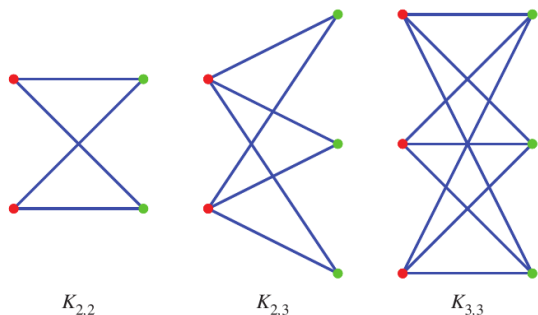
3

Notice that in our picture of K_4 there is an intersection right in the middle. This is not a vertex! It is simply an artifact of the picture. Can you find a picture of K_4 without any extra crossings like this? What about K_5 ?



4

2. The *complete bipartite graph* on $K_{m,n}$ has vertex set $V = V_1 \sqcup V_2$ where V_1 and V_2 have m and n elements, respectively. There is an edge for each pair $\{v, w\}$, where $v \in V_1$ and $w \in V_2$. If we think of the elements of V_1 as “red vertices” and the elements of V_2 as “green vertices,” then there are edges connecting every red vertex to every green vertex, and no edges connecting vertices of the same color. (More generally, a *bipartite graph* is one where there are “red” and “green” vertices and all edges connect vertices of different colors.)



5

3. Here’s an interesting example: consider the graph where the vertices are the two-element subsets of $\{1, 2, 3, 4, 5\}$ and where there is an edge between two

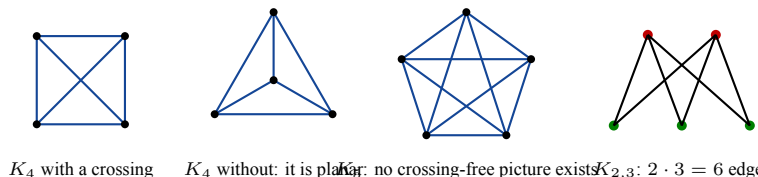
1 Figure 2.1

Check the claim “every vertex has valence 4” by counting half-edges at each dot. A useful identity: the sum of all valences equals twice the number of edges (each edge has two ends). So a graph with v vertices all of valence 4 has $2v$ edges.

2 The complete graph K_n

Every pair of distinct vertices is joined once, so K_n has $\binom{n}{2} = \frac{n(n-1)}{2}$ edges and every vertex has valence $n - 1$. K_1 is a dot, K_2 an edge, K_3 a triangle.

3 The crossing in K_4 , and the question about K_5



K_4 can be redrawn with no crossings (put one vertex in the middle of the triangle). K_5 cannot, whatever you do: this is a theorem (part of Kuratowski’s theorem, 1930). A graph drawable without crossings is called *planar*. The picture-crossing is “not a vertex”: the graph is the abstract data $(V, E, \text{endpoints})$; the drawing is only a way to look at it.

4 Complete bipartite graphs $K_{m,n}$

$m + n$ vertices in two colours, mn edges, every red vertex adjacent to every green one. $K_{3,3}$ is the other basic non-planar graph (the “three houses, three utilities” puzzle has no solution). $K_{1,n}$ is a star.

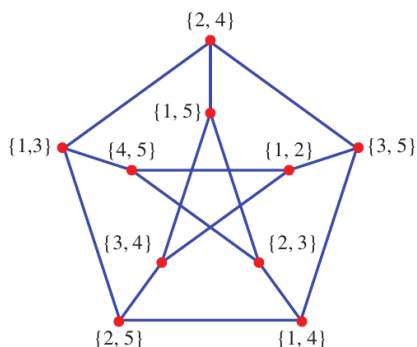
Try it

What are the valences in $K_{m,n}$? (Red vertices have valence n , green ones m .) Deduce again that there are mn edges.

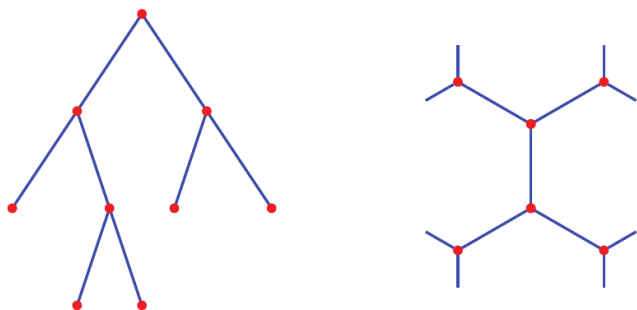
5 The Petersen graph, next page

Vertices are the $\binom{5}{2} = 10$ two-element subsets of $\{1, 2, 3, 4, 5\}$, edges join disjoint subsets. Visual page V2 draws it with every vertex labelled.

- 1 subsets if they are disjoint. This graph is known as the Petersen graph; it is a famous and important example in graph theory. Each vertex of the Petersen graph has valence 3. For example, the vertex $\{1, 2\}$ is adjacent to $\{3, 4\}$, $\{3, 5\}$, and $\{4, 5\}$.



- 2 4. A *tree* is a connected graph without a cycle. Finite trees are very important when studying searching and sorting algorithms in computer science. They also make an appearance in Sean Cleary's Office Hour 16 on Thompson's group. Infinite trees play an essential role in geometric group theory; see Office Hour 3. Below are two pictures of trees, the one on the right is a portion of T_3 , the tree in which every vertex has valence 3. Sometimes T_3 is called a regular 3-valent tree.



- 4 5. Graphs are commonly used to describe networks or to show complex relationships. For instance, we could create a graph whose vertices are the members of a social network (such as Facebook) and whose edges connect two members that are friends. (In some social networks, such as Google Plus, it is possible for friendships to be one-directional; to describe such a social network we need directed graphs, which we'll get to in a little bit.)

Here is a similar example: consider the graph that has as vertex set the set of movie actors and has edges between actors who have appeared in the

1 Petersen: counting

From $\{1, 2\}$ the disjoint pairs are the $\binom{3}{2} = 3$ two-element subsets of $\{3, 4, 5\}$, so every vertex has valence 3, and there are $10 \cdot 3/2 = 15$ edges. The standard drawing (outer pentagon, inner pentagram, five spokes) is on V2 with the subset labels.

Looking ahead

The Petersen graph is famous as the smallest counterexample to many plausible conjectures in graph theory. For us it matters as an example with a large symmetry group (S_5 , of order 120, page 27) that is not immediately visible in the picture.

2 Trees

Two words to define: a graph is *connected* if any two vertices are joined by a path of edges; a *cycle* is a closed path that does not repeat an edge (a loop or a double edge already counts as a cycle). A tree is connected and has no cycles. Consequences: between any two vertices of a tree there is exactly one path without backtracking, and a finite tree with v vertices has $v - 1$ edges.

Try it

Prove the " $v - 1$ edges" claim by induction: a finite tree with at least two vertices has a vertex of valence 1 (follow a path as far as it goes); remove it.

3 T_3 , the regular 3-valent tree

Every vertex has valence 3, so the tree is infinite. V2 draws a portion. Compare with the tree T_4 on page 32, the Cayley graph of F_2 .

Looking ahead

T_3 is also a Cayley graph: colour its edges with three colours so that each vertex sees one edge of each colour; then it is the Cayley graph of the group $\langle a, b, c \mid a^2, b^2, c^2 \rangle$, three reflections with no further relations. Office Hour 3 explains why groups acting freely on trees are "free" in this sense.

4 Networks

The Facebook graph and the actors graph are ordinary (undirected, unlabelled) graphs. "Bacon number" is the distance from Kevin Bacon in the path metric of page 36. One-directional friendships need directed graphs, defined on the next page.

Interlude: D. Pages 24–25: a graph is data, not a picture

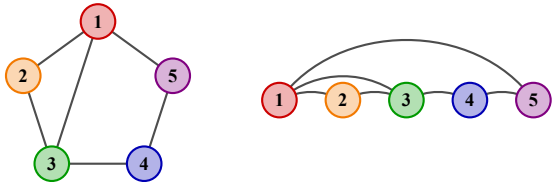
A graph is two sets: a set V of *vertices* and a set E of *edges*, each edge being a pair of vertices. That is all. A *drawing* is a choice of where to put the dots and how to route the lines, and the same graph has infinitely many drawings. Everything the book says about isomorphism, automorphism and Cayley graphs is about the data, never about the drawing.

One graph, two drawings

Both pictures below have $V = \{1, 2, 3, 4, 5\}$ and

$$E = \{\{1, 2\}, \{2, 3\}, \{3, 4\}, \{4, 5\}, \{5, 1\}, \{1, 3\}\},$$

so they are the same graph. Nothing you can read off one and not the other is a property of the graph.



drawing 1: a pentagon with one chord drawing 2: the same five vertices in a row, the same six edges as arcs

Watch out
A crossing in a drawing is not a vertex. The book makes this point with K_4 : it can be drawn with a crossing or without, and the crossing is an artifact of the drawing. K_5 cannot be drawn without one, which is a genuine fact about the graph, not about the drawing, and it is where graph theory starts to get interesting.

Why?
What *is* a property of the graph: the number of vertices, the number of edges, the *valence* of each vertex, meaning how many edges meet it, whether the graph is connected, and which vertices are joined to which. In the picture above every vertex has valence 2 except 1 and 3, which have valence 3. Count the ends of edges: each edge has two, so the valences always add up to twice the number of edges, here $2 + 3 + 3 + 2 + 2 = 12 = 2 \cdot 6$.

The families the book names

graph	vertices	edges	valences
K_n (complete)	n	$\frac{n(n-1)}{2}$	all $n - 1$
K_3	3	3	2
K_4	4	6	3
K_5	5	10	4
$K_{m,n}$ (bipartite)	$m + n$	mn	n and m
C_n (cycle)	n	n	all 2
Petersen	10	15	all 3
T_3 (the 3-valent tree)	infinite	infinite	all 3

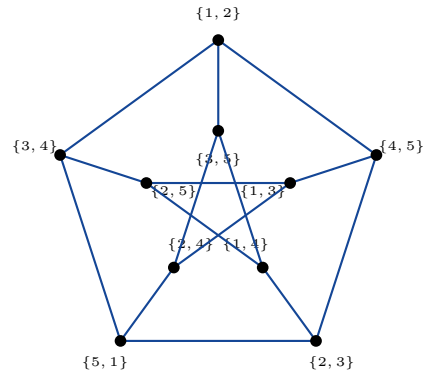
Why?
Why K_n has $n(n - 1)/2$ edges. There are n choices for one end and $n - 1$ for the other, but every edge has been counted twice, once from each end. The same bookkeeping gives the Petersen graph: 10 vertices of valence 3 make 30 edge-ends, hence 15 edges.

Watch out
The Petersen graph is drawn in the book three times, and the three pictures look nothing alike. They are the same graph: vertices are the ten two-element subsets of $\{1, \dots, 5\}$, joined when they are disjoint. Checking that a picture really is the Petersen graph means checking the data, for instance that $\{1, 2\}$ is joined to exactly $\{3, 4\}$, $\{3, 5\}$, $\{4, 5\}$.

Try it
Write down V and E for $K_{2,3}$ and check the edge count. Then draw C_6 twice, once as a hexagon and once with the six vertices on a line, and convince yourself the two drawings carry the same data.

Interlude: V2. A gallery of graphs

The Petersen graph, with its labels (page 25).

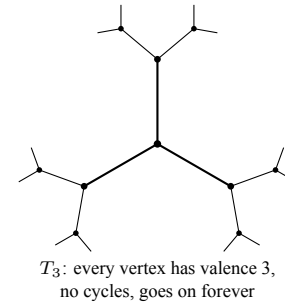


Outer pentagon: $\{1, 2\}, \{3, 4\}, \{5, 1\}, \{2, 3\}, \{4, 5\}$, consecutive ones disjoint. Each inner vertex is the unique two-element set disjoint from its outer neighbour and from the two outer sets next to it. Check a few edges: $\{1, 2\}$ is joined to $\{3, 4\}, \{4, 5\}$ (outer) and $\{3, 5\}$ (spoke): exactly the three subsets of $\{3, 4, 5\}$.

Try it

Permuting the numbers $1, \dots, 5$ permutes the labels and sends edges to edges: that is the action of S_5 on the Petersen graph (page 27). Find the permutation that rotates the outer pentagon one step. (It cycles the five outer labels, so try $(1\ 3\ 5\ 2\ 4)$ and check.)

A tree (page 25).



Formal definitions for Exercise 7 (page 26).

- *loopless*: the endpoint function never gives a pair $\{v, v\}$;
- *simple*: loopless and the endpoint function is injective (no two edges with the same endpoints);
- *directed*: the endpoint function lands in *ordered* pairs (v, w) : initial vertex v , terminal vertex w ;
- *labelled*: extra functions $V \rightarrow L$ and/or $E \rightarrow L$ into a set of labels L (letters, colours, generators).

A Cayley graph (page 29) is directed and edge-labelled, and it is simple whenever no generator equals its own inverse.

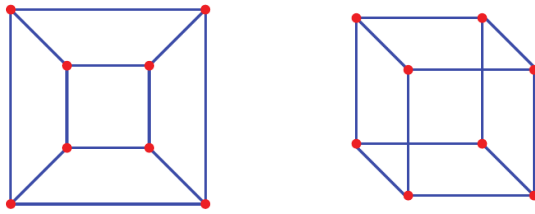


Figure 2.2 Two isomorphic graphs.

1

same movie. Extensive knowledge of this graph would be useful in playing the parlor game Six Degrees of Kevin Bacon.

2

Variations. In certain contexts, we might be interested in variations on the definition of a graph. For instance, in some cases we will care about *loopless graphs* (no edges connecting a vertex to itself) or *simple graphs* (at most one edge between two given vertices).

We also might be interested in *directed graphs*, where the edges have preferred directions, and in *labeled graphs*, where the vertices and/or edges come with a label (e.g., a letter or a color). When we draw a picture of a directed graph, we usually draw little arrows on the edges to indicate the preferred direction. We already encountered directed graphs in Office Hour 1 with edge labels when we constructed the geometric object that has $\mathbb{Z}^2$ as its group of symmetries.

Exercise 7. Give the formal definitions of simple graphs, loopless graphs, directed graphs, and labeled graphs.

3

Graph isomorphisms. You should have some intuitive idea of when two graphs are the same, for example, the two graphs in Figure 2.2. Formally, if Γ and Γ' are graphs with vertex sets V and V' and edge sets E and E' , then an *isomorphism* $\Gamma \rightarrow \Gamma'$ is a pair of bijections $V \rightarrow V'$ and $E \rightarrow E'$ that are compatible with the endpoint function in the sense that the endpoints of the image of an edge are the images of the endpoints of the edge. So just as an isomorphism of groups is a bijection that preserves all of the group structure, an isomorphism of graphs is a bijection (or rather a pair of bijections) that preserves all of the graph structure. For either groups or graphs, an isomorphism is a certificate that two a priori different objects are really the same object.

Sometimes the isomorphism is not clear, as in Figure 2.2. You might be surprised to learn that both graphs in Figure 2.3 are isomorphic to the Petersen graph!

4

Exercise 8. Find an explicit isomorphism between the graphs in Figure 2.2.

Exercise 9. Show that $K_{m,n}$ is isomorphic to $K_{n,m}$.

Exercise 10. Show that both graphs in Figure 2.3 are isomorphic to the Petersen graph.

1 Figure 2.2: two drawings of the same graph

Both graphs have 8 vertices, 12 edges and valence 3 everywhere. Both are the *cube graph*: the vertices and edges of a cube. Visual page V3 labels the vertices of both drawings by binary strings so that the isomorphism becomes visible (this is Exercise 8).

2 Variations

Loopless, simple, directed, labelled: V2 gives the formal definitions for Exercise 7. Directed labelled graphs are the ones Cayley graphs will be; the arrowed n -gon and the coloured grid of Office Hour 1 are examples.

3 Graph isomorphism

Two bijections, one on vertices and one on edges, such that the edge bijection sends the endpoints of e to the endpoints of the image of e . Compare the three “sameness” notions of the book side by side:

objects	isomorphism preserves	self-isomorphisms
groups	multiplication	automorphisms
graphs	endpoints (and labels, arrows)	$\text{Aut}(\Gamma)$, page 27
metric spaces	distances	$\text{Isom}(X)$, page 38

Why?

Why a *pair* of bijections rather than just one on vertices? Because of multiple edges: two vertices joined by three edges need the three edges matched individually. For simple graphs the vertex bijection determines everything.

4 Exercises 8–10: hints

(8) Use the binary labels of V3: an edge joins strings differing in one digit, in both drawings. (9) Swap the roles of the red and green sets; the edge set $\{v, w\}$ is unchanged as an unordered pair. (10) First check invariants: 10 vertices, 15 edges, valence 3, no cycles shorter than 5. Then label the vertices of each drawing of Figure 2.3 by two-element subsets so that adjacency means disjointness; the labels on V2 are a template.

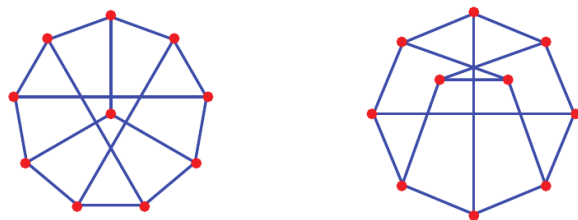


Figure 2.3 Two graphs isomorphic to the Petersen graph.

- 1 If we are working with graphs that have extra structure (i.e. directed and/or labeled edges) then an isomorphism must preserve this extra structure. For instance, if we have two graphs with blue and green edges, then an isomorphism should take blue edges to blue edges and green edges to green edges. Similarly, if we have directed graphs, then a graph isomorphism should respect the arrows on the edges.

- 2 **Graph automorphisms.** An isomorphism from a graph to itself is called an *automorphism*. The automorphisms of a graph are exactly what you should think of as its symmetries. The set of automorphisms of a graph Γ forms a group denoted by $\text{Aut}(\Gamma)$.

- 3 The graph on the left-hand side of Figure 2.2 has lots of obvious automorphisms, for instance, rotation by $\pi/2$ and reflection about the horizontal. It also has less-obvious automorphisms, for instance, ones that interchange the inner and outer squares.

Since the graph on the right-hand side of Figure 2.2 is isomorphic to the one on the left-hand side, it has the same group of automorphisms. What do the less-obvious automorphisms from the left-hand side look like on the right-hand side?

- 4 **Exercise 11.** Determine the automorphism group of the graph in Figure 2.2. *Hint: the discussion at the beginning of Office Hour 1 should help.*

We already have a number of examples of graphs. Here are some of their automorphisms:

- 5
1. Since each vertex in K_n is connected to every other vertex, all vertices are interchangeable; thus $\text{Aut}(K_n) \cong S_n$.
 2. Similarly, we find that if $m \neq n$, then $\text{Aut}(K_{m,n}) \cong S_m \times S_n$. What about $\text{Aut}(K_{n,n})$?
 3. From our original definition of the Petersen graph (using subsets of $\{1, 2, 3, 4, 5\}$) we see that S_5 is a subgroup of the automorphism group.
- 6
4. The regular 3-valent tree T_3 has lots of automorphisms. For instance, we can rotate by $2\pi/3$ at any vertex or swap two branches at a vertex (of course we need to be careful about how to define the automorphism on the rest of the tree). There are infinite order automorphisms as well. It turns out that $\text{Aut}(T_3)$ is uncountable!

1 Isomorphisms of decorated graphs

Same idea with more to preserve: colours go to the same colours, arrows to arrows pointing the same way. The more decoration, the fewer isomorphisms; this is used on page 31 to pin down the symmetries of a Cayley graph.

2 $\text{Aut}(\Gamma)$ is a group

The composition of two automorphisms preserves endpoints, hence is an automorphism; so is the inverse of one; the identity is one. Composition is associative because it is composition of functions. So $\text{Aut}(\Gamma)$ is a group under composition, exactly like the symmetries of the cube in Office Hour 1.

3 Obvious and less obvious automorphisms of the cube graph

Rotating the left-hand picture of Figure 2.2 by $\pi/2$, or reflecting it, gives 8 symmetries visible in the drawing. Swapping the inner and outer squares is invisible in the drawing but, on the cube, it is just a reflection of the cube in a plane (or, in the binary labels of V_3 , flipping the last digit). Exercise 11: the automorphism group is the full symmetry group of the cube, rotations and reflections, of order 48.

Why?

Why 48 and not more? An automorphism sends 000 to any of 8 vertices, then its three neighbours to the three neighbours of the image in any of $3! = 6$ orders; this determines everything. $8 \cdot 6 = 48$, and every such choice is realised by a symmetry of the solid cube.

4 $\text{Aut}(K_n) = S_n$

Every bijection of the vertices preserves edges because all pairs are edges.

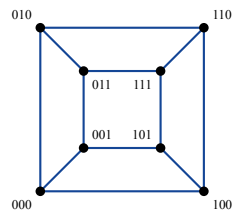
5 $\text{Aut}(K_{m,n})$

An automorphism must preserve the colour classes when $m \neq n$ (they have different sizes and valences), so it is a permutation of each class: $S_m \times S_n$, meaning pairs (σ, τ) composed componentwise. For $K_{n,n}$ one may also swap the two sides, giving $2 \cdot (n!)^2$ automorphisms.

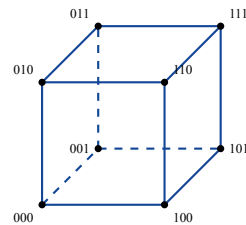
6 Petersen and T_3

S_5 acts on the Petersen graph through the labels (V2), and it is the whole automorphism group (Exercise 12: yes, $|\text{Aut}| = 120$). For T_3 : rotating by $2\pi/3$ about a vertex, or swapping two of its three branches, are automorphisms of finite order; an *infinite order* one slides the tree along a bi-infinite path (Exercise 13), like Φ_a on page 33.

Interlude: V3. One graph, two drawings, forty-eight symmetries



left graph of Figure 2.2



right graph of Figure 2.2 = the cube

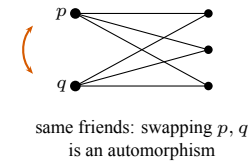
label every vertex by a binary string; two vertices are joined exactly when the strings differ in one digit.
The labels are the isomorphism (Exercise 8): same labels, same edges.

Reading the less obvious automorphisms (page 27). In the labels, the drawing's $\pi/2$ rotation is the map $xyz \mapsto \dots$ that cycles the outer square; inner-outer swap is $xyz \mapsto xy\bar{z}$ (flip the last digit). In general *any* permutation of the three digit positions combined with *any* pattern of digit flips is an automorphism: $3! \cdot 2^3 = 48$, all of them.

Looking ahead

The same labels give the n -dimensional cube graph Q_n : vertices are binary strings of length n , edges join strings differing in one digit. $\text{Aut}(Q_n)$ has order $2^n n!$. On page 37 the cube graph reappears as a metric space, and on page 38 the Hamming distance turns out to be its path metric.

Automorphisms from twins (page 28).



Exercise 14: other sources of Facebook automorphisms include any two *components* (groups of people with no connections between them) that happen to be isomorphic as graphs, and any two people p, q who are friends with each other and have the same other friends.

Try it

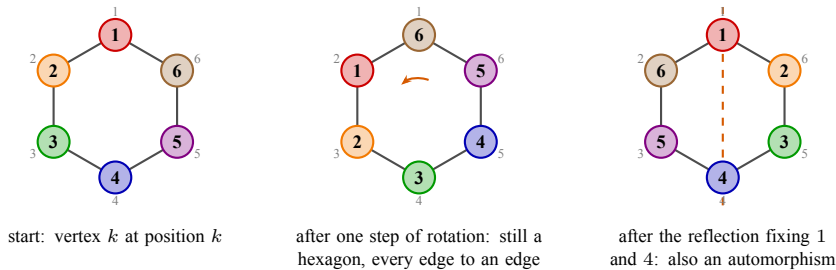
Exercise 12 sketch: an automorphism of the Petersen graph is determined by where it sends one vertex v (10 choices) and how it permutes the 3 neighbours of v (6 choices), and one shows the remaining 6 vertices are then forced: $10 \cdot 6 \cdot 2 = 120$ once a final binary choice is checked, and $|S_5| = 120$. So S_5 is everything.

Interlude: E. Pages 26–28: isomorphism relabels, automorphism is symmetry

An *isomorphism* between graphs is a bijection of the vertices that carries edges to edges and non-edges to non-edges. An *automorphism* is an isomorphism from a graph to itself, and the automorphisms form a group $\text{Aut}(\Gamma)$ under composition. So automorphism is the word “symmetry” made precise for graphs: no distances, no angles, only the question of which vertices are joined.

An automorphism, before and after

Positions $1, \dots, 6$ are fixed around a hexagon; a coloured disc names the vertex originally there. A picture is an automorphism exactly when, after the move, every pair of discs joined by an edge is still joined by an edge.



Why?

Why these are all of them. An automorphism of the hexagon must send 1 somewhere, six choices; and then its two neighbours must go to the two neighbours of the image, two choices; and after that everything is forced, because each further vertex is the unmarked neighbour of the previous one. That gives $6 \cdot 2 = 12$ automorphisms, and they are exactly the rotations and reflections: $\text{Aut}(C_6) = D_6$.

Watch out

An automorphism need not look like a rigid motion of the drawing. The book’s Figure 2.2 shows a graph drawn as two nested squares joined by spokes, and it has automorphisms that exchange the inner square with the outer one. Nothing in the drawing suggests them; they are visible only in the data, and they are still symmetries. This is the reason for insisting that a graph is not a picture.

Why?

Extra structure must be preserved too. If the edges carry colours, an isomorphism has to send each colour to the same colour; if they carry directions, it has to respect the arrows. This is what makes the count come out right for Cayley graphs: the arrowed n -gon has $2n$ symmetries as an undirected graph but only n once the arrows must be preserved, which is how $\mathbb{Z}/n\mathbb{Z}$ appeared in Office Hour 1.

graph	Aut	size
K_n	S_n	$n!$
C_n (cycle), $n \geq 3$	D_n	$2n$
the path on n vertices	$\mathbb{Z}/2\mathbb{Z}$	2
the n -gon with arrows one way round	$\mathbb{Z}/n\mathbb{Z}$	n
$K_{m,n}$, $m \neq n$	$S_m \times S_n$	$m! n!$
Petersen	S_5	120
T_3	infinite	—

Why?

Why $\text{Aut}(K_n) = S_n$. In a complete graph every pair of vertices is joined, so *every* bijection of the vertices carries edges to edges. There is nothing to preserve, so all $n!$ relabellings are automorphisms. At the other extreme, the more structure a graph has, the fewer automorphisms it admits, which is exactly the trick used to cut D_n down to $\mathbb{Z}/n\mathbb{Z}$.

Big picture

Automorphism groups are the supply of examples this office hour needs. Every graph hands you a group. The next question, and the point of the Cayley graph, is the converse: given a group, can you find a graph whose automorphism group is exactly that group? Page 29 answers yes, and the construction is uniform.

Try it

How many automorphisms has the graph consisting of two disjoint triangles? (Count the ways to permute within each triangle, and then whether the triangles can be exchanged.) Find an automorphism of the Petersen graph carrying $\{1, 2\}$ to $\{3, 4\}$, working with subsets rather than with the picture.

- 1 5. If two people have the same friends on Facebook, then there is an automorphism of the Facebook graph that interchanges these two people and fixes every other vertex (whether or not the two people are friends with each other!).

- 2 **Exercise 12.** Is S_5 the full group of automorphisms of the Petersen graph?

Exercise 13. Describe an infinite order element in $\text{Aut}(T_3)$.

Exercise 14. Besides the one mentioned above, can you think of other scenarios that lead to graph automorphisms of the Facebook graph?

- 3 If a graph Γ has extra structure (i.e., directed and/or labeled edges), then automorphisms of the graph are required to preserve this extra structure (after all, a graph automorphism is a special case of a graph isomorphism). The result is still a group and is still denoted by $\text{Aut}(\Gamma)$. For instance, in Office Hour 1 we considered automorphisms of a regular n -gon, thought of as a directed graph where the edges are all directed in the counterclockwise direction. As we saw, the group of automorphisms of this directed graph is $\mathbb{Z}/n\mathbb{Z}$.

- 4 **Group actions on graphs.** An *action* of a group G on a (possibly directed, possibly labeled) graph Γ is a homomorphism $G \rightarrow \text{Aut}(\Gamma)$. So it is based on the same idea as a group action, but an action on a graph needs to preserve the structure of the graph.

If we unravel the definition of a group action of G on an (unlabeled, undirected) graph Γ with vertex set V and edge set E , we have:

- 5
- for each $g \in G$ and $v \in V$ a choice of element of V , denoted by $g \cdot v$,
 - for each $g \in G$ and $e \in E$ a choice of element of E , denoted by $g \cdot e$,
 - for x in either V or E , we have $1 \cdot x = x$,
 - for $g, h \in G$ and x in V or E , we have $g \cdot (h \cdot x) = (gh) \cdot x$, and
 - if the endpoints of $e \in E$ are v and w , then the endpoints of $g \cdot e$ are $g \cdot v$ and $g \cdot w$.

In (yet) other words, a group action of G on Γ is a pair of group actions of G on V and E that are compatible with the endpoint function.

If we take Γ to be a labeled graph, then the second condition above would further stipulate that the labels on e and $g \cdot e$ need to be the same.

If we take Γ to be a directed graph, then the last condition above would say: if the initial and terminal endpoints of $e \in E$ are v and w , respectively, then the initial and terminal endpoints of $g \cdot e$ are $g \cdot v$ and $g \cdot w$, respectively.

We can rephrase our list of automorphisms of graphs in terms of group actions:

- 6
1. S_n acts on K_n by permuting the vertices
 2. $S_m \times S_n$ acts on $K_{m,n}$ by permuting the two sets of vertices
 3. S_5 acts on the Petersen graph via its action on $\{1, 2, 3, 4, 5\}$
 4. $\mathbb{Z}/3\mathbb{Z}$ acts on the regular 3-valent tree T_3 (as do $\mathbb{Z}/2\mathbb{Z}$ and $\mathbb{Z}$)
 5. $\mathbb{Z}/2\mathbb{Z}$ sometimes acts on the Facebook graph (for instance, when there are two people with the same set of friends, as above)

1 Twins

Two vertices with the same set of neighbours can be exchanged with no other change. V_3 draws it. This automorphism has order 2 and fixes everything else, so it generates a copy of $\mathbb{Z}/2\mathbb{Z}$ inside Aut .

2 Exercises 12–14

Hints are on V_3 . For 13, choose a path $\dots v_{-1}v_0v_1v_2\dots$ through T_3 with no backtracking and slide each v_i to v_{i+1} ; the branches hanging off the path are carried along. Repeating never returns a vertex to its place, so the order is infinite.

3 Decorations again

For a directed or labelled graph, “automorphism” means an isomorphism that preserves the decoration. This is how Office Hour 1’s arrowed n -gon ended up with only n symmetries: without the arrows it would have $2n$.

4 Action on a graph = homomorphism $G \rightarrow \text{Aut}(\Gamma)$

The same pattern as page 22 with S_X replaced by $\text{Aut}(\Gamma)$. Unravelling: an action on the vertex set, an action on the edge set, the two compatible with the endpoint function.

Why?

Why is compatibility the right condition? Because that is exactly what makes each map $x \mapsto g \cdot x$ an automorphism: it must send the endpoints of e to the endpoints of $g \cdot e$. Axioms 1 and 2 then say $g \mapsto (\text{this automorphism})$ is a homomorphism.

5 Directed and labelled versions

For labels: $g \cdot e$ carries the same label as e . For arrows: initial goes to initial, terminal to terminal. Both will be needed for Cayley graphs.

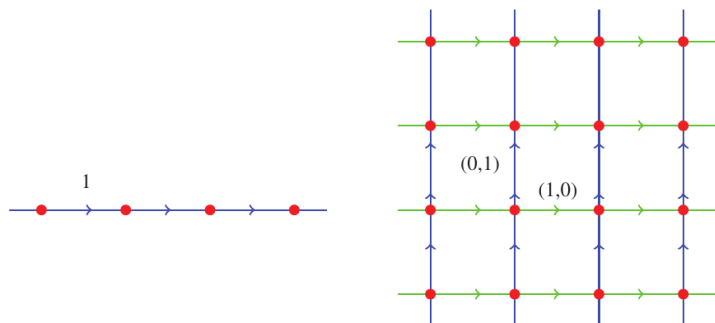
6 The five examples as actions

(1) S_n on K_n ; (2) $S_m \times S_n$ on $K_{m,n}$; (3) S_5 on Petersen via labels; (4) $\mathbb{Z}/3\mathbb{Z}$ rotates T_3 about a vertex, $\mathbb{Z}/2\mathbb{Z}$ swaps two branches, $\mathbb{Z}$ slides along a path; (5) $\mathbb{Z}/2\mathbb{Z}$ swaps twins.

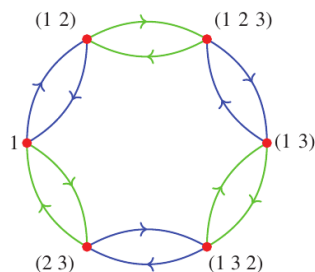
Looking ahead

The next page finally constructs the graph on which G acts “by itself”: its Cayley graph. Everything on pages 23–28 was preparation for stating precisely what “ G is the symmetry group of this graph” means.

- 1 Cayley graphs.** Finally, we come to the whole reason we have been studying graphs. Let G be an arbitrary group and let S be a generating set for G . For simplicity we assume that S does not contain the identity. The *Cayley graph* for G with respect to S is a directed, labeled graph $\Gamma(G, S)$ given as follows: the vertex set is G , there is a directed edge from g to gs for every $g \in G$ and $s \in S$, and we label the edge from g to gs by the element s of G . (If S is small we can sometimes use colors for the edges instead of labels.)
- 2** You might wonder what happens if you take two different generating sets for the same group. How are the two Cayley graphs related? They might not be isomorphic (for instance, if the cardinalities of the generating sets are different). However, there is a sense in which they are the same; see Office Hour 7 on quasi-isometries for an explanation.
- 3** The Cayley graphs for $\mathbb{Z}$ with respect to $\{1\}$ and for $\mathbb{Z}^2$ with respect to $\{(1, 0), (0, 1)\}$ are familiar. Indeed, we already discovered them in Office Hour 1! Each blue edge in the graph for $\mathbb{Z}$ corresponds to the generator $\{1\}$. The green and blue edges in the graph for $\mathbb{Z}^2$ correspond to the generators $(1, 0)$ and $(0, 1)$, respectively. (Of course, 1 here is the number 1, not the identity.)



- 4** Here is the Cayley graph for S_3 with respect to the generating set $\{(1\ 2), (2\ 3)\}$. The blue edges correspond to the generator $(1\ 2)$ and the green edges correspond to the generator $(2\ 3)$.



1 The Cayley graph $\Gamma(G, S)$

Vertices: the elements of G . Edges: from g to gs , labelled s , for each generator s . So every vertex has exactly one outgoing edge and one incoming edge per generator, hence valence $2|S|$ (page 32).

Watch out

The edge goes from g to gs , generator on the *right*. This is not a matter of taste: it is what makes left multiplication $v \mapsto hv$ a symmetry (page 30), because $h(gs) = (hg)s$. With edges $g \rightarrow sg$ instead, the symmetries would be right multiplications.

Why?

Why exclude the identity from S ? An edge from g to $g \cdot 1 = g$ would be a loop at every vertex and carry no information. And when some s satisfies $s = s^{-1}$, the two edges $g \rightarrow gs$ and $gs \rightarrow g$ point opposite ways between the same vertices; they are usually drawn as one undirected edge, as in the S_3 picture below.

2 Different generating sets

Same group, different graphs. $\mathbb{Z}$ with $\{1\}$ is a line; with $\{2, 3\}$ it is the graph on V_4 , which looks different close up but is still “a line” from far away. Office Hour 7 makes “from far away” precise (quasi-isometry) and proves that the large-scale shape does not depend on S .

3 $\mathbb{Z}$ and $\mathbb{Z}^2$

The arrowed line and the coloured grid from Office Hour 1, pages 10–11, were Cayley graphs all along.

4 S_3 with $\{(1\ 2), (2\ 3)\}$

Walk around the hexagon from 1 multiplying on the right: $1 \rightarrow (1\ 2) \rightarrow (1\ 2)(2\ 3) = (1\ 2\ 3) \rightarrow (1\ 2\ 3)(1\ 2) = (1\ 3) \rightarrow (1\ 3)(2\ 3) = (1\ 3\ 2) \rightarrow (1\ 3\ 2)(1\ 2) = (2\ 3) \rightarrow (2\ 3)(2\ 3) = 1$. Six steps, alternating colours, back to the start: the hexagon encodes the relation $((1\ 2)(2\ 3))^3 = 1$.

Try it

Check one of these products with the table method of Office Hour 1, page 7, remembering that the right-hand factor acts first.



Figure 2.4 Part of a Cayley graph.

- 1 Exercise 15.** Draw the Cayley graph for $\mathbb{Z}$ with the generating set $\{2, 3\}$, for D_4 with the generating set $\{s, t\}$, for S_3 with the generating set $\{(1\ 2), (1\ 2\ 3)\}$, and for S_4 with the generating set $\{(1\ 2), (2\ 3), (3\ 4)\}$.

Here's a useful observation about the Cayley graph made evident by Figure 2.4:

- 2** If $s_1, s_2, \dots, s_n$ are the labels (in order) on an edge path in $\Gamma(G, S)$ from 1 to g , then $g = s_1s_2 \cdots s_n$. And vice versa, if $g = s_1s_2 \cdots s_n$, then there is an edge path from 1 to g in $\Gamma(G, S)$ whose labels are $s_1, s_2, \dots, s_n$ (in that order).

(We allow s_i to be the inverse of an element s of S , meaning that our path crosses an s -edge against the direction of the edge.) So there is a correspondence between paths in the Cayley graph from 1 to g on one hand and words in the generators representing g on the other hand.

- 3 Exercise 16.** What do the labels on an edge path from g to h in $\Gamma(G, S)$ tell you about g and h ?

- 4 Automorphisms of the Cayley graph.** Each element of a group G gives rise to an automorphism of the Cayley graph $\Gamma(G, S)$, as follows. The automorphism Φ_g associated to $g \in G$ is given on the vertices by:

$$\Phi_g(v) = gv.$$

This is just the action of G on itself by left multiplication. Since there is at most one directed edge connecting any two vertices with a specified direction, and since no edges connect a vertex to itself, there is only one way to extend this to an automorphism of the entire directed, labeled graph: Φ_g sends the edge $v \rightarrow vs$ to the edge $gv \rightarrow gvs$. This works since $\Phi_g(v) = gv$ and $\Phi_g(vs) = gvs$.

- 5 THEOREM 2.2.** Let S be a generating set for G . Then the map $G \rightarrow \text{Aut}(\Gamma(G, S))$ is an isomorphism.

- The proof requires one new idea. Let S^{-1} denote the set of inverses of elements of S . The *word length* of $g \in G$ with respect to S is the length of the shortest word in $S \cup S^{-1}$ that is equal to g in G . The identity $1 \in G$ has word length 0. By the definition of a generating set, every element of G has finite word length with respect to S .

Our earlier observation about edge paths in Cayley graphs shows that we can also think of word length geometrically: the word length of g is the fewest number of edges in an edge path from 1 to g .

1 Exercise 15

All four graphs are drawn or described on V4: $\mathbb{Z}$ with $\{2, 3\}$, D_4 with the two reflections, S_3 with a transposition and a 3-cycle, and S_4 with the three adjacent transpositions (24 vertices; its shape is the *permutohedron*, a solid with 6 square and 8 hexagonal faces).

2 Paths are words

The dictionary of the chapter. Starting at 1 and following edges labelled $s_1, s_2, \dots, s_n$ in order, you arrive at the vertex $s_1s_2 \cdots s_n$, because each step multiplies on the right by the label. Following an s -edge *backwards* multiplies by s^{-1} . Hence: paths from 1 to $g \leftrightarrow$ words in $S \cup S^{-1}$ equal to g . A closed path from 1 back to 1 is a word equal to the identity, a *relator* in the sense of Office Hour 1, page 18.

3 Exercise 16

A path from g to h with labels $s_1, \dots, s_n$ means $h = gs_1 \cdots s_n$, i.e. the labels spell the element $g^{-1}h$. This is the formula behind the word metric on page 37.

4 The automorphism Φ_g : left multiplication

On vertices $v \mapsto gv$; on the edge $v \rightarrow vs$, it gives $gv \rightarrow gvs$, which is again an s -edge. So Φ_g preserves labels and directions. The paragraph's remark "at most one directed edge, no loops" is what makes the edge map forced once the vertex map is known.

5 Theorem 2.2: the promised theorem

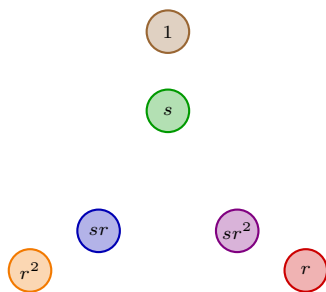
The map $g \mapsto \Phi_g$ is an isomorphism from G onto *all* label-preserving automorphisms of the Cayley graph. Every group is the full symmetry group of a graph. Proof on the next page; V5 draws its key step.

6 Word length

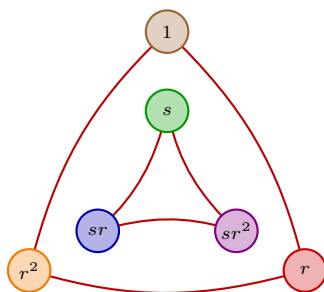
$|g|_S$ = length of the shortest word in the generators and their inverses equal to g . Geometrically: the smallest number of edges in a path from 1 to g . Examples: in $\mathbb{Z}$ with $\{1\}$, $|n| = |n|$; with $\{2, 3\}$, $|1| = 2$ (as $3 \cdot 2^{-1}$). In $\mathbb{Z}^2$ with the standard generators, $|(m, n)| = |m| + |n|$: V4 draws the balls of radius 1, 2, 3, and they are diamonds.

Interlude: F. Pages 29–30: building a Cayley graph in three steps

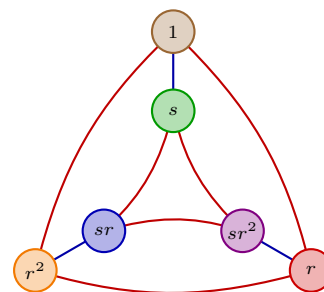
The recipe has no choices in it once G and a generating set S are fixed: the vertices are the elements of G , and for each element g and each generator s there is an edge from g to gs , labelled by s . Here $G = S_3 = D_3$ and $S = \{r, s\}$ with r of order three and s of order two, and the graph is assembled one generator at a time.



step 1 one vertex per group element, six in all



step 2 a red arrow $g \rightarrow gr$ for the generator r : two triangles, wound opposite ways



step 3 a blue edge $g \rightarrow gs$ for the generator s ; since $s^2 = 1$ it needs no arrowhead

Why?

Why the two triangles wind in opposite directions. The outer triangle is $1 \rightarrow r \rightarrow r^2 \rightarrow 1$. The inner one is $s \rightarrow sr \rightarrow sr^2 \rightarrow s$, and because s is a reflection it reverses the sense of rotation: as drawn, with each inner vertex placed beside its outer partner, the inner arrows run the other way round. That is the picture of the relation $srs = r^{-1}$, and it is visible in the graph without any algebra.

Why?

Why s -edges need no arrowhead. An arrow should run from g to gs and back from gs to $gs^2 = g$, so the two directions coincide whenever $s^2 = 1$. The convention is to draw such a pair as a single undirected edge. For a generator of infinite order, like 1 in $\mathbb{Z}$, the arrowhead carries real information.

Watch out

The edges use right multiplication. The edge out of g labelled s goes to gs , not to sg . This is not a matter of taste. In the next pages the group acts on the graph by *left* multiplication, $x \mapsto gx$, and the reason that action carries edges to edges is the associativity computation $(gx)s = g(xs)$: multiplying on the left commutes with multiplying on the right. Had the edges been built with left multiplication, the action would have had to be on the right, and the same picture would appear with every arrow reversed.

G	S	the graph	valence
$\mathbb{Z}$	$\{1\}$	the arrowed line	2
$\mathbb{Z}$	$\{2, 3\}$	a line with two edge colours	4
$\mathbb{Z}/n\mathbb{Z}$	$\{1\}$	the arrowed n -gon	2
$\mathbb{Z}^2$	$\{(1, 0), (0, 1)\}$	the coloured grid	4
F_2	$\{a, b\}$	the 4-valent tree	4
S_3	$\{r, s\}$	two triangles plus three spokes	3
D_4	$\{s, t\}$, two reflections	an octagon, two colours	2
S_4	the three neighbour swaps	the permutohedron	3

Why?

Why the valence is $2|S|$, with a correction. At each vertex g there is one edge out for each generator, to gs , and one edge in for each generator, from gs^{-1} . That is $2|S|$ ends, unless a generator has order two, when the outgoing and incoming edges for it coincide and the count drops by one for that generator. S_3 with $\{r, s\}$ therefore has valence $2 + 1 = 3$.

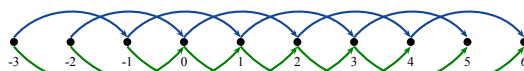
Watch out

Different generating sets give different graphs for the same group, and they need not even have the same valence: compare the two rows for $\mathbb{Z}$ in the table. So the Cayley graph is a graph attached to a *pair* (G, S) . Office Hour 7 explains the sense in which all of them look the same from far away.

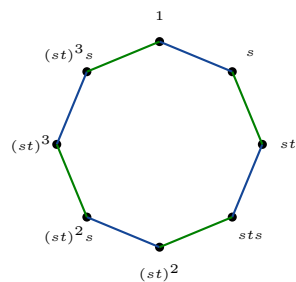
Try it

This is Exercise 15. Draw the Cayley graph of $\mathbb{Z}$ with $S = \{2, 3\}$, of D_4 with two reflections, of S_3 with $\{(1\ 2), (1\ 2\ 3)\}$, and of S_4 with the three neighbour swaps. For the third, check your picture against the one above, since $(1\ 2\ 3)$ is r and $(1\ 2)$ is a reflection.

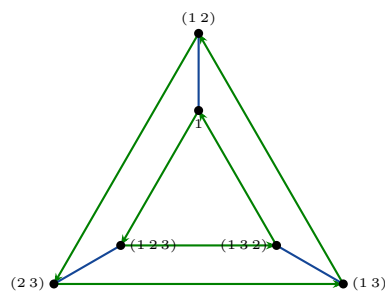
Interlude: V4. Cayley graphs for Exercise 15, and balls in $\mathbb{Z}^2$



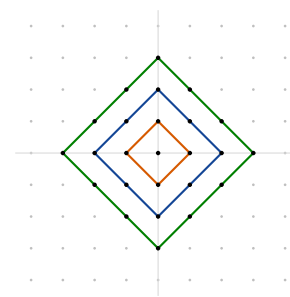
$\mathbb{Z}$ with $S = \{2, 3\}$: blue = $+2$, green = $+3$.
Word length of 1 is 2 (as $3 - 2$), of 5 is 2 ($2 + 3$), of 6 is 2 ($3 + 3$), of 7 is 3.



D_4 with the two reflections s (blue), t (green):
an 8-cycle with alternating colours; $(st)^4 = 1$ closes it



S_3 with $\{(12), (123)\}$: green arrows = (123) ,
blue = (12) (two-way). A triangular prism.



$\mathbb{Z}^2$: elements of word length $\leq 1, 2, 3$.
The spheres are diamonds: the taxicab metric.

Why?

Why does S_3 with $\{(12), (123)\}$ give a prism rather than a hexagon? The 3-cycle has order 3, so its edges form triangles (the cosets of the subgroup $\{1, (123), (132)\}$); the transposition has order 2 and its edges pair the two triangles up. Different generating sets, different pictures, same group: compare with the hexagon on page 29.

Try it

S_4 with $\{(12), (23), (34)\}$ has 24 vertices and 36 edges. Start drawing: from 1 the three edges lead to the three transpositions; (12) and (34) commute, so the four elements $1, (12), (34), (12)(34)$ form a square; (12) and (23) satisfy the braid relation, so they generate a hexagon. Squares and hexagons fit together into a truncated octahedron.

Big picture

The diamonds on the right are the reason for the phrase “the word metric on $\mathbb{Z}^2$ is the taxicab metric” on page 37. Word length counts steps along grid edges, and the set of points reachable in at most r steps is the taxicab ball of radius r . Counting the points in these balls, $1, 5, 13, 25, \dots$ (growing like $2r^2$), is the start of Office Hour 12 on growth of groups.

Looking ahead

For F_2 the ball of radius r in T_4 has $1 + 4 + 12 + \dots + 4 \cdot 3^{r-1}$ vertices, growing exponentially. Polynomial versus exponential growth is a property of the group, not of the generators, and it distinguishes $\mathbb{Z}^2$ from F_2 at a glance.

- 1 Let's prove Theorem 2.2. To simplify notation, let $\Gamma = \Gamma(G, S)$. It is not hard to see that the map $G \rightarrow \text{Aut}(\Gamma)$ we have already defined is a homomorphism. It is injective since different elements of G send the identity vertex of Γ to different vertices. The main thing is to show that it is surjective, that is, every element of $\text{Aut}(\Gamma)$ is equal to some Φ_g .

Let $\Phi \in \text{Aut}(\Gamma)$, that is, Φ is an isomorphism $\Gamma \rightarrow \Gamma$, where Γ is regarded as a directed graph with edge labels. Suppose that $\Phi(1) = g$ (we are using the same symbol for a vertex and the corresponding element of Γ). We will show that $\Phi = \Phi_g$. As above, since directed edges are determined by their endpoints, it is enough to show that Φ agrees with Φ_g on the vertices of Γ .

- 2 We use induction on word length in G with respect to S . The base case is word length 0, which is just the statement $\Phi(1) = g = \Phi_g(1)$. Assume Φ agrees with Φ_g on all elements of G of word length n with respect to S . Suppose $v \in G$ has word length $n + 1$. This means that $v = ws$, where the word length of w is n and $s \in S \cup S^{-1}$. To simplify matters, assume $s \in S$; the other case is similar.

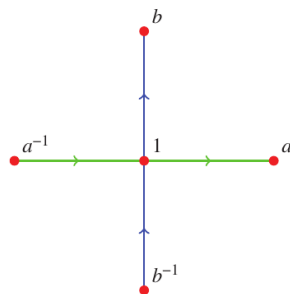
- 3 By assumption, $\Phi(w) = \Phi_g(w)$. There is a unique edge labeled s emanating from w , namely, the edge from w to $ws = v$. Similarly, there is a unique edge labeled s emanating from $\Phi(w)$, with ending point $\Phi(w)s$. Since Φ and Φ_g both respect edge labels, it must be that $\Phi(v) = \Phi_g(v) = \Phi(w)s$. In particular, $\Phi(v) = \Phi_g(v)$, which is what we wanted to show!

- 4 **The Cayley graph for the free group.** By Theorem 2.2, we know there is a geometric object with F_2 as its group of symmetries, namely, the Cayley graph for F_2 (with respect to any generating set). But what does such a Cayley graph look like?

Let a and b be the standard generators for F_2 . We can draw a picture of the Cayley graph $\Gamma = \Gamma(F_2, \{a, b\})$ as follows. We start by drawing the vertex for the identity element 1 of F_2 (the empty word):

1

- 5 The four edges containing 1 connect 1 to the vertices a, b, a^{-1} , and b^{-1} , the four elements of F_2 with word length 1:



1 Structure of the proof

Three things: (a) $g \mapsto \Phi_g$ is a homomorphism, because $\Phi_g \Phi_h(v) = ghv = \Phi_{gh}(v)$; (b) it is injective, because $\Phi_g(1) = g$ recovers g ; (c) it is surjective: every label-preserving automorphism Φ equals some Φ_g . Only (c) needs an idea.

2 The idea: induction on word length

Set $g = \Phi(1)$; the claim is $\Phi = \Phi_g$. Both agree at 1. Suppose they agree at w and let $v = ws$ be one step further. There is exactly one s -edge leaving w and exactly one leaving $\Phi(w)$; an automorphism preserving labels and directions must match them. So $\Phi(v)$ is forced to be $\Phi(w)s = gws = \Phi_g(v)$. Since every vertex is reachable from 1 in finitely many steps (S generates), agreement spreads to the whole graph. V5 draws the step.

Why?

Why does the proof need labels and arrows? Because “exactly one s -edge leaving w ” is what pins $\Phi(v)$ down. Without labels the graph may have extra symmetries not coming from the group: V5 shows the hexagon of $\mathbb{Z}/6\mathbb{Z}$, whose undirected version has 12 automorphisms.

3 The case $s \in S^{-1}$

If $v = ws^{-1}$, use the unique s -edge arriving at w instead. Same argument.

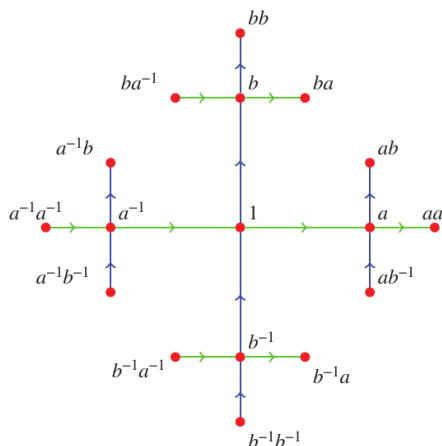
4 The Cayley graph of F_2 , stage by stage

Start at 1; the four neighbours are the four reduced words of length 1. Each of them has three new neighbours (the fourth edge goes back to 1). Because reduced words are distinct group elements, no vertex is ever drawn twice, and the graph is the tree T_4 of page 32.

5 Word length 1

The vertices a, b, a^{-1}, b^{-1} at distance 1. In general the vertices at distance n from 1 are the reduced words of length n , and there are $4 \cdot 3^{n-1}$ of them.

What are the edges incident to these newly drawn vertices? In general, the number of edges incident to a vertex is twice the cardinality of S , so there will be three new edges at each vertex:



- 1 At the n th stage of this process, we will have $4(3^{n-1})$ new vertices, and we will have to draw $4(3^n)$ new edges. Since the label of each new vertex is a reduced word in a and b , and since distinct reduced words give distinct elements of F_2 , we will never see a repeated vertex. Thus, the Cayley graph of F_2 is a tree. In fact it is T_4 , the tree in which every vertex has valence 4.

Let's think about how elements of F_2 give symmetries of this Cayley graph. The first key thing to remember is that all edges look the same from the point of view of an abstract graph even though we have drawn them with different lengths in our picture.

- 3 To present a concrete example, we'll consider the element a , and its symmetry Φ_a of the Cayley graph. First let's think about what Φ_a does to the vertices, i.e., reduced words in a and b . Multiplying a word that starts with a on the left by a results in a word that starts aa . Likewise, multiplying a word that starts with b or b^{-1} on the left by a results in a word that starts ab or ab^{-1} , respectively. What about words that start with a^{-1} ? Well, it depends on the second letter; this will be the new first letter. So the options are a^{-1} , b , and b^{-1} . We cannot have a as the new first letter since that would have meant that our initial reduced word started $a^{-1}a$, which means it was not reduced.

- 4 To see the action on Γ it is useful to divide the vertices of the Cayley graph Γ into four regions, corresponding the first letter of the corresponding words. There is the a region Γ_a , the b region Γ_b , the a^{-1} region $\Gamma_{a^{-1}}$, and the b^{-1} region $\Gamma_{b^{-1}}$ (we can take 1 to be a fifth region on its own):

1 T_4 , and why there is no repeated vertex

"Distinct reduced words give distinct elements of F_2 " is the uniqueness of reduced form from Office Hour 1 (Exercise 5 there). If some vertex appeared twice, two different reduced words would name the same element. So the Cayley graph is a tree, and since each vertex has valence $2|S| = 4$ it is T_4 .

Try it

Count: 4 vertices at distance 1, 12 at distance 2, 36 at distance 3. Check $4 \cdot 3^{n-1}$ against the picture on the page for $n = 2$.

2 "All edges look the same"

The drawing must shrink edges to fit infinitely many into a finite picture; the abstract graph has no notion of length yet (that comes with the path metric, page 36, where every edge has length 1). An automorphism may send a "long" drawn edge to a "short" one; nothing is wrong.

3 What Φ_a does to words

Left multiplication by a , then reduce:

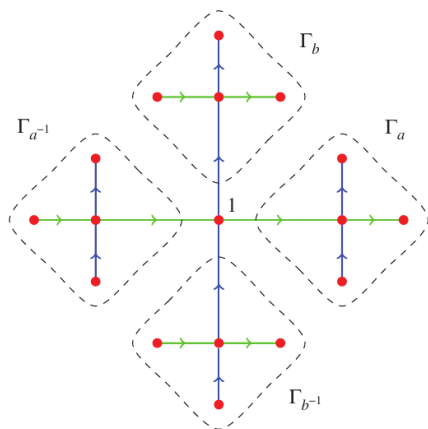
word w starts with	$a \cdot w$ starts with
a	aa
b or b^{-1}	ab or ab^{-1}
a^{-1}	the second letter of w (a^{-1} , b or b^{-1})

The last row is the interesting one: $a \cdot a^{-1}w' = w'$, so the first letter is cancelled and the word gets *shorter*. In the tree, $a^{-1}w'$ sits in the left branch and is pulled to w' , one step closer to the centre or across it.

4 The four regions

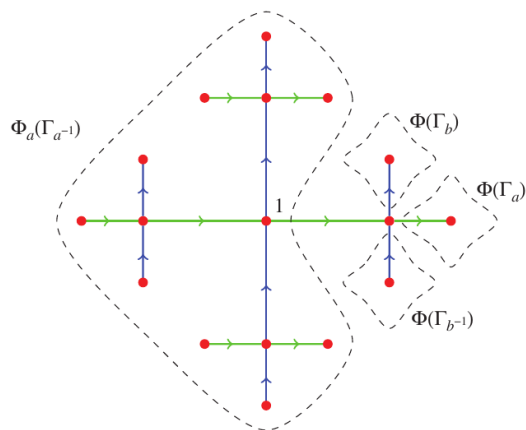
$\Gamma_a, \Gamma_b, \Gamma_{a^{-1}}, \Gamma_{b^{-1}}$ are the four branches hanging off 1, one per first letter. Every vertex except 1 lies in exactly one of them. The next page pictures Φ_a region by region; V5 shows the same map as a translation along an axis.

1



2

Using these regions, we can give the following rough description of what Φ_a does to the vertices of Γ : it takes $\Gamma_{a^{-1}}$ onto $\Gamma_{a^{-1}} \cup \Gamma_{b^{-1}} \cup \Gamma_b \cup \{1\}$. It also takes 1 to a , and $\Gamma_a \cup \Gamma_b \cup \Gamma_{b^{-1}} \cup \{1\}$ onto Γ_a .



3

It might look like $\Phi_a(\Gamma_{a^{-1}})$ is three times the size of $\Gamma_{a^{-1}}$. However, you should convince yourself that these two (infinite) subgraphs of the Cayley graph are isomorphic as graphs. The distortion is just an artifact of our picture.

4

Try to describe the picture of Φ_w for more complicated w . And read Office Hour 3 for a much deeper discussion of what these symmetries can look like.

1 The regions

The branch Γ_a contains all reduced words beginning with a : $a, aa, ab, ab^{-1}, aaa, \dots$. It is itself a tree in which the root a has 3 children and every other vertex has 3 children.

2 Φ_a region by region

From the table on page 32: words beginning with a, b or b^{-1} , and the word 1, all acquire the prefix a , so $\Gamma_a \cup \Gamma_b \cup \Gamma_{b^{-1}} \cup \{1\}$ is mapped *into* Γ_a ; and words beginning with a^{-1} lose that letter, so $\Gamma_{a^{-1}}$ is mapped onto everything else. Both maps are bijections onto their targets because $\Phi_{a^{-1}}$ undoes them.

Why?

Why does the picture look like a distortion? Φ_a is really a *translation*: it slides the whole tree one edge along the horizontal axis $\dots, a^{-2}, a^{-1}, 1, a, a^2, \dots$ (V5). Everything to the left of 1 moves right by one edge and stays a tree of the same shape; the drawing only makes the moved copy look three times bigger because the page has finite room.

3 The two subgraphs are isomorphic

$\Gamma_{a^{-1}}$ is a tree with a root (a^{-1}) of valence 3 (inside the region) and all other vertices of valence 3 in the region except for the one edge to the root. The union of $\Gamma_{a^{-1}}, \Gamma_b, \Gamma_{b^{-1}}$ and $\{1\}$ is a tree with root 1 of valence 3, its neighbours being a^{-1}, b, b^{-1} , and the same branching pattern. Same shape, so isomorphic, and Φ_a is an explicit isomorphism.

4 More complicated w

Φ_{a^2} slides two steps along the same axis. Φ_{ab} slides along the axis running through the powers of ab , namely $\dots, (ab)^{-1}, 1, ab, (ab)^2, \dots$, which is a zigzag line in the tree. $\Phi_{bab^{-1}}$ is a conjugate of Φ_a and slides along the axis through b : the translation axis has been moved by Φ_b .

Looking ahead

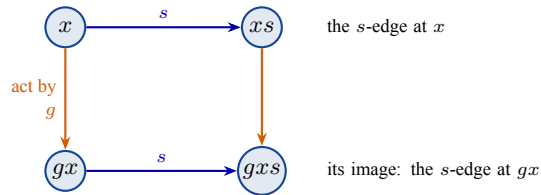
Office Hour 3 proves that every nontrivial element of a group acting freely on a tree is such a translation along an axis, and uses this to show that such groups are free.

Interlude: G. Pages 31–33: why the symmetries of the Cayley graph are exactly G

Theorem 2.2 says that for $\Gamma = \Gamma(G, S)$, regarded as a graph with directed, labelled edges, $\text{Aut}(\Gamma) \cong G$. The proof has three separate jobs, and it is worth separating them: the map is defined, it is injective, it is surjective. Write ρ_g for the relabelling $x \mapsto gx$ of the vertices.

1. Each ρ_g really is an automorphism

An automorphism has to send edges to edges, keeping labels and directions. The s -labelled edges are the pairs $x \rightarrow xs$, so the question is whether $gx \rightarrow gxs$ is again such a pair, and it is, because $(gx)s = g(xs)$.



Big picture

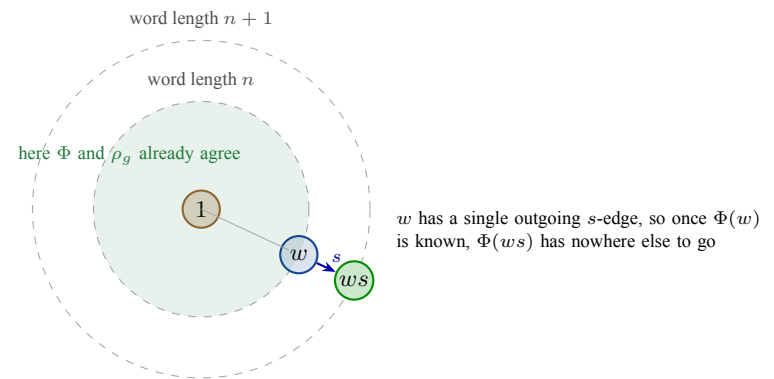
That single line, $(gx)s = g(xs)$, is the whole reason the construction works, and it is why the edges were built with multiplication on the *right* while the group acts on the *left*. Left and right multiplication commute with each other, so each leaves the other's structure intact. If both had been on the same side, the action would have destroyed the edges.

2. Injective: an automorphism is pinned down by where 1 goes

If $\rho_g = \rho_h$ then applying both to the vertex 1 gives $g = h$. So distinct group elements give distinct automorphisms, and G sits inside $\text{Aut}(\Gamma)$.

3. Surjective: nothing else is a symmetry

Let Φ be any automorphism and let $g = \Phi(1)$. The claim is $\Phi = \rho_g$, and the proof is an induction on word length. Both maps agree at 1, which is the case $n = 0$. Suppose they agree on every element of word length n , and let v have word length $n + 1$. Then $v = ws$ for some w of length n and some generator s or inverse of one.



Since Φ preserves labels and directions, it must send the unique s -edge leaving w to the unique s -edge leaving $\Phi(w) = gw$. That edge ends at $gws = gv$, so $\Phi(v) = gv = \rho_g(v)$, completing the induction.

Why?

Where each hypothesis is used. *S generates G* is what makes every vertex reachable, so the induction covers everything. *Labels* are what force the s -edge to go to an s -edge. *Directions* are what prevent Φ from sending the edge out of w to the edge coming into it. Drop any one and the theorem fails.

Watch out

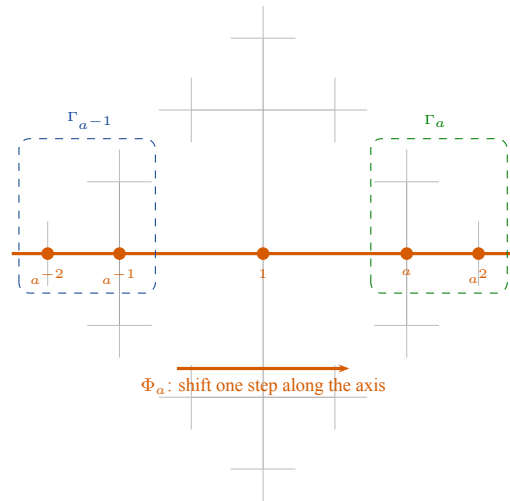
Forgetting the extra structure changes the answer. The hexagon with arrows one way round has 6 symmetries, matching $\mathbb{Z}/6\mathbb{Z}$; the same hexagon with the arrows rubbed out has 12, namely D_6 . The Cayley graph of $\mathbb{Z}^2$ without its two edge colours has extra symmetries too, the quarter turns of Office Hour 1. Theorem 2.2 is a statement about the decorated graph.

Try it

Take $\Gamma(S_3, \{r, s\})$ from the previous interlude. Check by hand that ρ_s exchanges the two triangles and that it preserves the red arrows. Then find an automorphism of the underlying *undirected, unlabelled* graph that is not any ρ_g , and see exactly which condition it violates.

Interlude: V5. Two pictures for Theorem 2.2

Φ_a is a translation (pages 32–33).

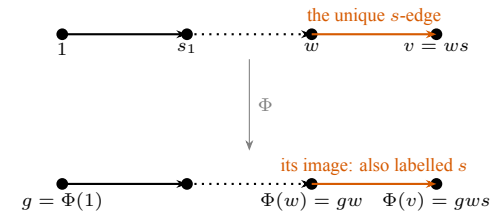


The orange line is the *axis* of a : the vertices a^n , $n \in \mathbb{Z}$. Φ_a moves each a^n to a^{n+1} and drags the branches along. The dashed blue region $\Gamma_{a^{-1}}$ is carried onto the whole left half of the tree together with the vertical branches at 1; the green region Γ_a is carried into itself, one step deeper.

Try it

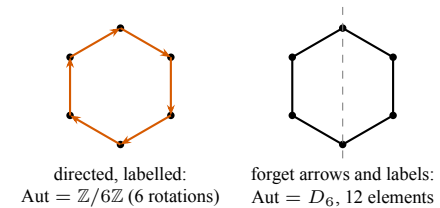
Where does Φ_a send the vertex $a^{-1}b$? (Answer: to b .) And $a^{-1}a^{-1}$? (To a^{-1} .) And b ? (To ab .) Locate all six vertices in the picture.

The inductive step of the proof (page 31).



Top row: a shortest path from 1 to v . Bottom row: its image under an arbitrary label-preserving automorphism Φ with $\Phi(1) = g$. Once $\Phi(w)$ is known, $\Phi(v)$ is forced, because it must be the end of the unique s -edge leaving $\Phi(w)$.

Why the labels matter.



Without arrows and labels, the reflections of the hexagon are automorphisms too, and Aut is twice as large as the group. Theorem 2.2 is about the *decorated* Cayley graph.

2.2 METRIC SPACES

- 1 We have solved the problem of finding a geometric object whose set of symmetries is a given group G . If we really want to understand a group G , we should try to understand an even bigger variety of geometric objects for which G is a set of symmetries. We'll now introduce metric spaces—a fairly general notion of a geometric object—and then we'll say what it means for G to be the collection of symmetries of a metric space. The idea is to mimic (rather, generalize) the notion of a group action on a graph. At the end we will state an observation that forms the cornerstone of geometric group theory: the group G can itself be thought of as a metric space, that is, G is itself a geometric object.

Metric spaces. In short, a metric space is a set with a way of measuring the distance between two points. You know lots of examples of these. The most basic example is $\mathbb{R}$, with the distance between x and y being $|x - y|$ (or more generally, the Euclidean metric on $\mathbb{R}^n$; see below). But there are other, more abstract, examples. For instance, the set could be all of the subway stations in a city and the “distance” could be the average transit time between those stations. We'll give more examples after we state the definition.

- 2 A *metric space* is a set X together with a function

$$d: X \times X \rightarrow \mathbb{R}$$

satisfying the following properties:

- *positive definiteness:* $d(x, y) \geq 0$ for all $x, y \in X$ and $d(x, y) = 0$ if and only if $x = y$
- *symmetry:* $d(x, y) = d(y, x)$ for all $x, y \in X$
- *triangle inequality:* $d(x, z) \leq d(x, y) + d(y, z)$ for all $x, y, z \in X$

The function d is called a *metric*, and is also called a *distance function*. Let's consider some examples.

- 3 1. *The Euclidean metric.* A metric space you are very familiar with is Euclidean n -space, that is, $\mathbb{R}^n$ with the Euclidean distance:

$$d((a_1, \dots, a_n), (b_1, \dots, b_n)) = \sqrt{(a_1 - b_1)^2 + \dots + (a_n - b_n)^2}.$$

- 4 2. *The taxicab metric.* Consider $\mathbb{R}^2$ with the following metric:

$$d((x_1, y_1), (x_2, y_2)) = |x_1 - x_2| + |y_1 - y_2|.$$

This distance is the (Euclidean) length of any shortest path from (x_1, y_1) to (x_2, y_2) that uses only horizontal and vertical line segments. The taxicab metric models the distance that a taxi must travel in a city whose streets are laid out in a rectangular grid.

1 Why go beyond graphs

The Cayley graph solves the “one object per group” problem. But a group can act on many spaces, and each action may reveal something (Theorem 2.3 on page 40 is the first example). Metric spaces are the general setting; the section ends with the observation that G itself is one.

2 The three axioms of a metric

Positive definiteness: distances are nonnegative, and zero only from a point to itself. Symmetry: same distance both ways. Triangle inequality: a detour through y is never shorter than going straight from x to z .

Background you need

On the real line $d(x, y) = |x - y|$. The triangle inequality there is $|a + b| \leq |a| + |b|$ with $a = x - y$, $b = y - z$. Absolute value is the one-dimensional case of everything below.

Try it

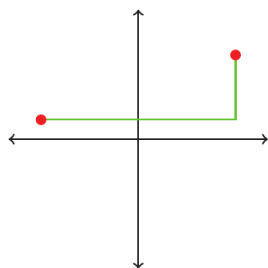
Non-examples, to see the axioms bite: $d(x, y) = (x - y)^2$ on $\mathbb{R}$ fails the triangle inequality ($d(0, 2) = 4 > 1 + 1$). Average transit time between subway stations is a metric only if the trip is equally fast in both directions. “Distance” along one-way streets is not symmetric.

3 The Euclidean metric

Pythagoras in n dimensions. The triangle inequality is genuinely a theorem here (it follows from the Cauchy–Schwarz inequality); the book takes it as known.

4 The taxicab metric

Horizontal plus vertical displacement. V6 draws the unit circle (a diamond) and shows that shortest paths are far from unique. This is the metric that will describe $\mathbb{Z}^2$ as a group (page 37).



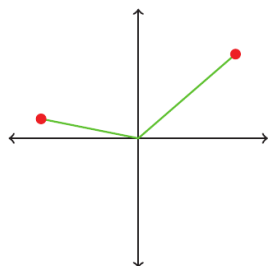
The taxicab metric makes sense in $\mathbb{R}^n$ as well:

$$d((a_1, \dots, a_n), (b_1, \dots, b_n)) = |a_1 - b_1| + \dots + |a_n - b_n|.$$

- 1 3. *The hub metric.* Consider $\mathbb{R}^2$ in polar coordinates (r, θ) . Assume $r_1, r_2 \geq 0$ and $\theta_1, \theta_2 \in [0, 2\pi)$. We define a metric by:

$$d((r_1, \theta_1), (r_2, \theta_2)) = \begin{cases} |r_1 - r_2| & \text{if } \theta_1 = \theta_2 \\ r_1 + r_2 & \text{otherwise.} \end{cases}$$

- 2 In other words, the distance between two points is the length of the shortest path, where all paths must travel along rays from the origin. This is reminiscent of a railway system where all rail lines pass through a single hub.



- 3 4. *The sup metric.* Consider $\mathbb{R}^2$ with the following metric:

$$d((x_1, y_1), (x_2, y_2)) = \max\{|x_1 - x_2|, |y_1 - y_2|\}.$$

A form of this metric is used by the U.S. Postal Service, which defines the length of a rectangular package to be the largest of its three dimensions.

- 4 5. *The L^1 norm.* Here is an example that is more abstract in nature. Let $C([0, 1])$ denote the set of continuous functions from $[0, 1]$ to $\mathbb{R}$. We define a metric on $C([0, 1])$ by:

$$d(f, g) = \int_0^1 |f(x) - g(x)| dx.$$

- 5 6. *The Hamming distance.* Here is an example that arises in coding theory. Let X denote the set of binary strings with n digits, that is, strings of 0's or 1's

1 The hub metric

Every journey runs along rays through the origin (the hub). Two points on the same ray are at their ordinary distance; otherwise you must go in to the hub and out again, $r_1 + r_2$. This is often called the “British Rail metric” or “SNCF metric” (all lines through London or Paris). V6 draws it.

Why?

Triangle inequality check, the only nontrivial case: x, z on the same ray, y on another. Then $d(x, z) = |r_x - r_z| \leq r_x + r_z = d(x, y) - r_y + d(y, z) - r_y \leq d(x, y) + d(y, z)$. The other cases are the triangle inequality on a line, or immediate.

2 Shortest paths through the hub

The book’s picture: the metric is a *path* metric, distance = length of the shortest allowed route, like the taxicab metric and the path metric on graphs. Many important metrics in this book are of this type.

3 The sup metric

The larger of the two coordinate differences. Its unit circle is a square (V6).

Big picture

On $\mathbb{R}^2$ the three metrics are within a factor 2 of one another: $d_{\text{sup}} \leq d_{\text{Eucl}} \leq d_{\text{taxi}} \leq 2 d_{\text{sup}}$. They disagree about the shapes of circles but agree about what is “far” and what is “near”. This is the first hint of the “roughly isometric” idea of Office Hour 7.

4 The L^1 distance between functions

Distance between two functions = area between their graphs (V6). Positive definiteness uses continuity: a continuous function with zero integral of $|f - g|$ must equal g everywhere.

5 Hamming distance

Defined on the next page: count the positions where two strings differ. It is the path metric on the cube graph (page 38 and V3).

that have length n . Given two strings s_1 and s_2 we define $d(s_1, s_2)$ to be the number of digits in which s_1 and s_2 differ. For example, if $n = 7$ we have

$$d(0110101, 1110000) = 3$$

as the two given strings differ in the first, fifth, and seventh digits.

- 2 7. *The path metric on a graph.* Now, for perhaps the most important example on this list (at least for us), let Γ be a *connected* graph, that is, a graph where there is an edge path between any two vertices. We can impose a metric on the set of vertices of Γ : the distance between two vertices is the length of the shortest edge path (that is, sequence of edges) connecting the vertices.

We'll see this metric a lot in the book. We'll refer to it as the *path metric* on a graph. We saw this idea when we defined the word length as the length of the shortest edge path from the identity to a given group element in the Cayley graph.

- 3 8. *The geometric realization of a graph.* The path metric on a graph is really a metric on the vertices of a graph. In particular, all distances are integers. But when you think about a graph you probably think of it as a continuous space, where it makes sense to move a small distance ϵ away from a vertex. Let's turn this intuition into reality.

The *geometric realization* X of a graph $\Gamma = \Gamma(V, E)$ is the space formed as follows. We start with one point for each element of V and one copy of the unit interval $[0, 1]$ for each element of E . Then we form X by identifying the endpoints of each interval with the V -points according to the endpoint function; in other words, as a set, X is the quotient of a disjoint union of intervals and points. The images of elements of V and E in the quotient are called the vertices and edges of X .

- 4 There is now a natural metric on X : for instance, if x and y are two points on the same edge of X (and assuming that the endpoints of the edge are not identified), then since we identified the edge with $[0, 1]$, the expression $|x - y|$ makes sense, and that is the distance between x and y . We'll leave it to you to define the distance between two points that do not lie on the same edge. Project 6 in Office Hour 3 asks you to fill in the details of this construction.

On an informal level you should think of Γ and X as being the same, but in practice there are situations where it is important to distinguish the one you are talking about from the other. Intuitively, we should think of Γ as a list of instructions for building a metric space and X as the finished product.

- 5 **Exercise 17.** Check that all of the above distance functions indeed define metrics on their respective spaces.
- 6 **Exercise 18.** Draw the unit sphere around the origin (that is, the set of points that have distance 1 from the origin) in the Euclidean metric, the taxicab metric, the hub metric, and the sup metric on $\mathbb{R}^2$.
- 7 **Exercise 19.** In which of the above spaces is there a unique shortest path between two points?

1 Hamming distance as a path

$0110101 \rightarrow 1110101 \rightarrow 1110001 \rightarrow 1110000$: each step changes one digit, and no shorter route exists since three digits differ. So the Hamming distance is the path metric on the graph whose vertices are strings and whose edges join strings differing in one digit: the cube graph Q_n of $V3$.

2 The path metric on a graph

Distance = fewest edges in a path. Connectedness guarantees the distance is finite. Triangle inequality: put a shortest path from x to y after a shortest path from y to z ; this is *some* path from x to z , so the shortest one is no longer.

Big picture

On a Cayley graph, the path distance from 1 to g is the word length of g (page 30). The path metric is the bridge between the combinatorial Cayley graph and the metric-space view of a group on page 37.

3 Geometric realisation

Replace each edge by a copy of $[0, 1]$ glued at its ends. Now there are points in the middle of edges and every distance can be any nonnegative real number, not only an integer. V6 draws the difference.

Why?

The distance the book leaves to you: for x inside edge e and y inside edge e' , try all four ways of leaving e at an endpoint u and entering e' at an endpoint u' , and take the smallest of $|x - u| + d(u, u') + |u' - y|$.

4 Instructions versus product

Γ is data (sets and a function); X is a space. In practice the book says “the graph” for both and lets context decide.

5 Exercise 17

The hub metric is the only case with something to check (annotation 1 on page 35). For the path metric use the concatenation argument in 2.

6 Exercise 18: the trap

Euclidean: circle. Taxicab: diamond $|x| + |y| = 1$. Sup: square $\max(|x|, |y|) = 1$. Hub, around the origin: the ordinary circle, since $d(0, p) = r$; but around any other point the “sphere” is two points on the same ray (V6). Always ask “around which point?”.

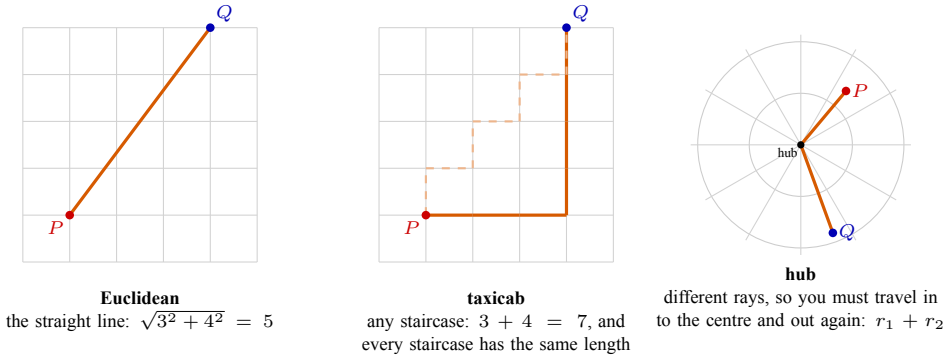
7 Exercise 19

Unique shortest path: Euclidean (the segment), hub (through the hub or along the ray). Not unique: taxicab and sup (any monotone staircase, V6), and the path metric on most graphs (two vertices opposite each other on a cycle).

Interlude: H. Pages 34–36: the three axioms for a distance, and five rulers that satisfy them

A metric space is a set X together with a function d that assigns a number to each pair of points, subject to three promises: $d(x, y) \geq 0$ with equality exactly when $x = y$; $d(x, y) = d(y, x)$; and the triangle inequality $d(x, z) \leq d(x, y) + d(y, z)$. Only the third does any real work. The same set carries many different metrics, and the metric, not the set, is what decides the geometry.

The same two points, measured three ways



Why?

Why the taxicab distance does not depend on the staircase. Any path made of unit steps from P to Q must make at least 3 horizontal steps and at least 4 vertical ones, and a shortest path never wastes a step by going backwards. So every monotone staircase has length 7, and there are many shortest paths, unlike the Euclidean case where there is exactly one. Having many geodesics is typical of the metrics that come from groups.

Why?

Why the triangle inequality holds for the hub metric. To get from P to Q you may either stay on one ray, when the distance is $|r_1 - r_2|$, or change rays, when you must pass through the hub and pay $r_1 + r_2$. Inserting a third point can only add another trip in and out, never shorten one, so the inequality survives. The picture also shows the price of the metric: two points a millimetre apart on different rays can be very far apart.

Candidate rulers, tested

set	rule $d(x, y)$	zero iff =	symmetric	triangle
$\mathbb{R}$	$ x - y $	✓	✓	✓
$\mathbb{R}^2$	Euclidean	✓	✓	✓
$\mathbb{R}^2$	taxicab	✓	✓	✓
$\mathbb{R}^2$	hub	✓	✓	✓
strings	digits that differ	✓	✓	✓
graph	shortest edge path	✓	✓	✓
$\mathbb{R}$	$(x - y)^2$	✓	✓	✗
$\mathbb{R}$	$ x + y $	✗	✓	✓
$\mathbb{R}$	$x - y$	✗	✗	✓
any	0 if $x = y$, else 1	✓	✓	✓

Why?

Why $(x - y)^2$ fails. Take $x = 0, y = 1, z = 2$. Then $d(x, z) = 4$ while $d(x, y) + d(y, z) = 1 + 1 = 2$, so the direct route is longer than the detour. Squaring exaggerates large distances, and the triangle inequality is exactly the promise that detours never help.

Watch out

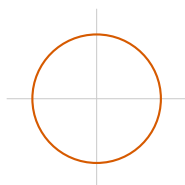
The first axiom has two halves and both matter. $d(x, y) = |x| + |y|$ is symmetric and satisfies the triangle inequality, but $d(x, x) = 2|x|$ is not zero, so distinct points cannot be told from equal ones. The last row of the table, the discrete metric, satisfies everything and yet carries no geometry at all: every distinct pair is at distance 1.

Try it

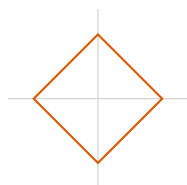
Check the three axioms for the taxicab metric directly from the formula $|a_1 - b_1| + |a_2 - b_2|$. Then decide whether $\sqrt{|x - y|}$ is a metric on $\mathbb{R}$, and whether $\min(|x - y|, 1)$ is.

Interlude: V6. Metric spaces you can draw

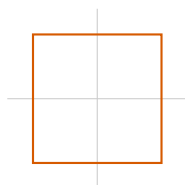
Unit spheres, i.e. all points at distance exactly 1 from a chosen point (Exercise 18).



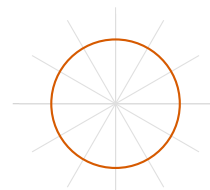
Euclidean: circle



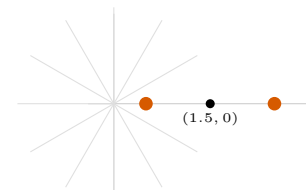
taxicab: diamond



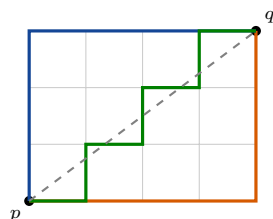
sup: square



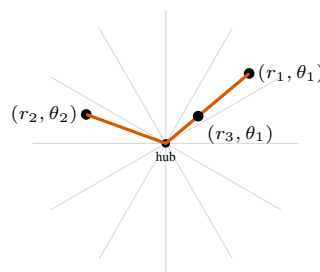
hub, around the origin:
a circle (every ray is a rail line)



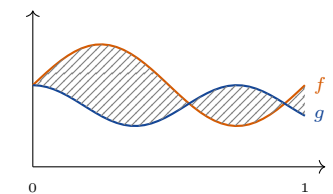
hub, around (1.5, 0):
just two points!



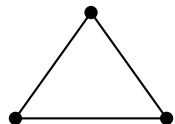
taxicab: many shortest paths, all of length $4 + 3 = 7$;
the dashed straight line has Euclidean length 5



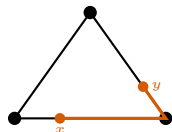
hub metric: to change ray you must pass the hub,
 $d = r_1 + r_2$; on the same ray $d = |r_1 - r_3|$



$d(f, g) = \int_0^1 |f - g|$ = the shaded area
between the two graphs



graph Γ : 3 vertices, 3 edges;
path metric: all distances 1



realisation X : points inside edges too;
 $d(x, y) = 0.7 + 0.25$ along the shortest route

Why?

Why is the taxicab sphere a diamond? $|x| + |y| = 1$ is four line segments, one per quadrant: in the first quadrant it is $x + y = 1$. The sup sphere $\max(|x|, |y|) = 1$ is the square with corners $(\pm 1, \pm 1)$. The Euclidean sphere lies between them, touching the diamond at the axes and the square at the corners: the picture of $d_{\text{sup}} \leq d_{\text{Eucl}} \leq d_{\text{taxi}}$.

Try it

In the taxicab plane, how many shortest paths run from $(0, 0)$ to $(4, 3)$ along grid lines? (Choose which 4 of the 7 steps go right: $\binom{7}{4} = 35$.) In the sup metric the straight segment is shortest, but so are many others.

- 1 A subset of a metric space is also a metric space using the same distance functions. So, for instance, we can consider a regular n -gon in $\mathbb{R}^2$ as a metric space with the Euclidean distance.

Groups as metric spaces. We now come to the most important examples of metric spaces in this book: groups!

- 2 Let G be a group and S a generating set. We declare the distance between $g, h \in G$ to be the word length of $g^{-1}h$ with respect to S . In particular, the distance between g and 1 is the word length of g . It is straightforward to see that this rule does indeed define a metric on G . We call it the *word metric*.

The word metric is the same as the path metric on the Cayley graph of G (thought of as an unlabeled, undirected graph): the distance between two group elements is the length of the shortest edge path between the corresponding vertices of the Cayley graph.

- 3 Here are two key examples: the word metric on $\mathbb{Z}$ with the generating set $\{1\}$ is simply the usual Euclidean metric that $\mathbb{Z}$ inherits as a subset of $\mathbb{R}$; the word metric on $\mathbb{Z}^2$ with the generating set $\{(1, 0), (0, 1)\}$ is the taxicab metric.

In summary, the word metric gives us:

- 4 A group together with a generating set is a metric space.

This is one of the important themes of geometric group theory. Some people might even say that the study of groups as metric spaces *is* geometric group theory.

Once we have turned a group into a geometric object (that is, a metric space), we can start to ask interesting questions about what a group looks like. There are many ways to interpret this: Timothy Riley's Office Hour 8 discusses isoperimetric problems for a group (what is the largest area enclosed by a loop of length ℓ ?), Moon Duchin's Office Hour 9 discusses groups that look roughly like hyperbolic spaces, Nic Koban and John Meier's Office Hour 10 talks about what a group can look like at infinity, Greg Bell's Office Hour 11 discusses the dimension of a group, and Eric Freeden's Office Hour 12 studies the number of group elements in a ball of radius r around the identity.

Exercise 20. Write down the word metrics for your favorite finite groups (with your favorite generating sets) by listing all distances explicitly.

- 5 **Isometries between metric spaces.** Let X and Y be metric spaces with distance functions d_X and d_Y . An *isometry* from X to Y is a bijective function $f: X \rightarrow Y$ so that

$$d_Y(f(x), f(y)) = d_X(x, y)$$

for all pairs $x, y \in X$. In other words, an isometry is a bijective function that preserves distances. So an isometry is to metric spaces what an isomorphism is to groups or graphs.

- 6 Here is an example. Let X be the set of vertices of the cube graph from Figure 2.2 and let d_X be the path metric on the graph. Let Y be the set of binary strings of length 3 and let d_Y be the Hamming distance. We claim that X and Y

1 Subsets inherit the metric

Restrict d to the subset; the three axioms still hold. So a polygon, a lattice $\mathbb{Z}^2 \subset \mathbb{R}^2$, or the set of vertices of a cube are metric spaces without further work.

2 The word metric

$d(g, h) = |g^{-1}h|_S$: the length of the shortest word taking you from g to h (Exercise 16, page 30). In particular $d(1, g) = |g|_S$.

Why?

Axioms: $d(g, h) = 0$ iff $g^{-1}h = 1$ iff $g = h$. Symmetry: $h^{-1}g = (g^{-1}h)^{-1}$, and a word and its inverse have the same length. Triangle: $g^{-1}k = (g^{-1}h)(h^{-1}k)$, so concatenating shortest words for the two factors gives a word for $g^{-1}k$ of length $d(g, h) + d(h, k)$.

Watch out

Why $g^{-1}h$ and not gh^{-1} ? Because with $g^{-1}h$ the metric is *left-invariant*: $d(kg, kh) = |(kg)^{-1}kh| = |g^{-1}h| = d(g, h)$. So left multiplication, the action of Theorem 2.2, is an isometry (page 39). With gh^{-1} one would get right-invariance instead, the wrong side for this book's conventions.

3 $\mathbb{Z}$ is the line, $\mathbb{Z}^2$ is the taxicab plane

$d(m, n) = |m - n|$ in $\mathbb{Z}$; $d((a, b), (c, d)) = |a - c| + |b - d|$ in $\mathbb{Z}^2$. The diamonds of V4 are the balls of this metric.

4 The slogan

A group with a generating set is a metric space. From here on a group has a *shape*, and one can ask what it looks like: Office Hours 8–12 each ask one such question (area of loops, hyperbolicity, ends, dimension, growth).

Looking ahead

The shape depends on the generating set only up to “rough” distortion (Office Hour 7), so the questions in those office hours have answers that belong to the group itself.

5 Isometries

A bijection that preserves all distances: the metric-space version of “isomorphism”. Injectivity is automatic (if $f(x) = f(y)$ then $d(x, y) = d(f(x), f(y)) = 0$), so only surjectivity needs checking.

Try it

Exercise 20 for D_3 with generators s, t (two reflections): the Cayley graph is the hexagon of page 29, so distances are hexagon distances, at most 3. V7 tabulates them.

6 The cube, three ways

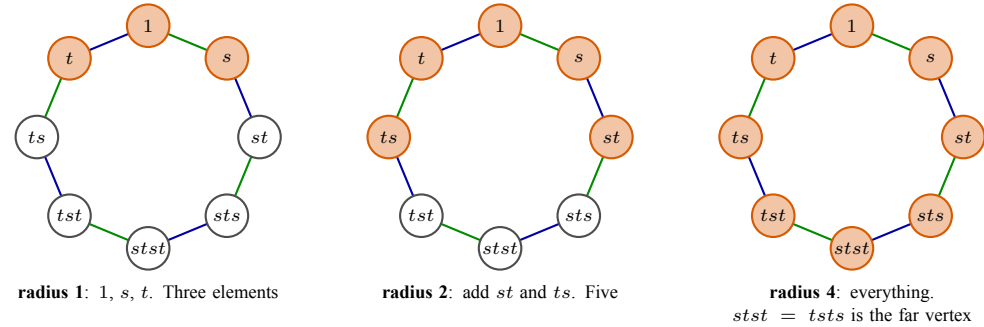
Vertices of the cube graph with the path metric, binary strings with Hamming distance, and $\{0, 1\}^3 \subset \mathbb{R}^3$ with the taxicab metric. The binary labels of V3 identify all three. Continued on the next page.

Interlude: J. Page 37: the word metric, and why a group is itself a geometric object

Fix a group G and a generating set S . The **word length** $|g|$ is the least number of letters from $S \cup S^{-1}$ needed to spell g , and the **word metric** is $d(g, h) = |g^{-1}h|$. In particular $d(1, g) = |g|$. This is the same thing as the path metric on the Cayley graph with its labels and arrows forgotten: the shortest spelling of $g^{-1}h$ is the shortest edge path from g to h .

Balls in D_4 with two reflections as generators

The Cayley graph is an octagon whose edges alternate between the two generators. Shaded vertices are those within the stated distance of the identity.



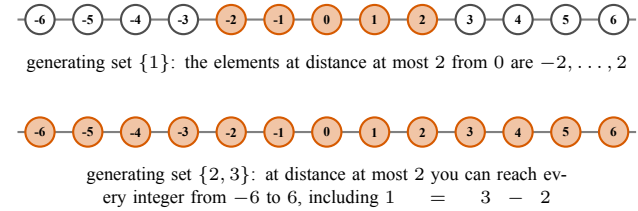
Why?

Why $d(g, h) = |g^{-1}h|$ is a metric. It is zero exactly when $g^{-1}h = 1$, that is when $g = h$. It is symmetric because $|x^{-1}| = |x|$: reverse the word and invert each letter. And the triangle inequality is concatenation: a spelling of $g^{-1}h$ followed by a spelling of $h^{-1}k$ spells $g^{-1}k$, so the shortest spelling of $g^{-1}k$ is no longer than the sum.

Why?

Why the formula uses $g^{-1}h$ and not gh^{-1} . The choice makes the metric **left invariant**: $d(kg, kh) = |(kg)^{-1}(kh)| = |g^{-1}k^{-1}kh| = |g^{-1}h| = d(g, h)$. So G acts on itself by isometries, which is exactly the statement that the group is a geometric object on which it acts. This is the cornerstone sentence of the office hour.

The metric depends on the generating set



Watch out

Same group, two different distances. With $S = \{1\}$ the integer 6 is at distance 6 from 0; with $S = \{2, 3\}$ it is at distance 2, since $6 = 3 + 3$. And 1 is at distance 1 in the first and distance 2 in the second, because $1 = 3 - 2$ needs two letters. So the word metric belongs to the pair (G, S) , not to G alone. Office Hour 7 shows that for finite generating sets the differences are bounded, which is the sense in which the geometry of the group is well defined after all.

G	S	$ g $ for a typical g	balls grow like
$\mathbb{Z}$	$\{1\}$	$ n $	$2k + 1$
$\mathbb{Z}^2$	two unit slides	$ m + n $	$2k^2 + 2k + 1$
F_2	$\{a, b\}$	the reduced length	$\approx 3^k$
D_4	two reflections	0, 1, 2, 3, 4	the octagon

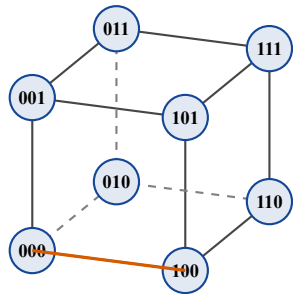
Try it

In the octagon above, read off $d(s, tst)$ by counting edges, then check it against $|s^{-1} \cdot tst| = |stst|$. In $\mathbb{Z}^2$ with the two unit slides, draw the ball of radius 3 and count its points; the shape is the taxicab diamond of the previous interlude.

Interlude: I. Pages 36–38: the path metric, and one metric space wearing three disguises

The example that matters for the rest of the book is the *path metric* on a connected graph: the distance between two vertices is the number of edges in a shortest edge path joining them. It satisfies the axioms for the same reasons the taxicab metric does, and a shortest path is called a *geodesic*. The pages then show that three spaces which look unrelated are the same metric space.

The cube graph, with its vertices named by strings



one edge = one digit changed = distance 1

An edge of the cube joins two corners differing in exactly one coordinate, and two binary strings at Hamming distance 1 differ in exactly one digit. So naming the corner (x, y, z) by the string xyz turns edges into pairs at distance 1, and hence shortest paths into shortest paths.

0	1	1	0	1	0	1
1	1	1	0	0	0	0
x			x		x	

they differ in three places, so the distance is 3

Why?

Why counting differing digits is a metric. It is clearly zero only for equal strings and clearly symmetric. For the triangle inequality, if s_1 and s_3 differ in a digit then at least one of s_1, s_2 or s_2, s_3 must differ there too, so every place counted on the left is counted at least once on the right.

The dictionary, in full

corner of the cube	binary string	coordinates
X : graph vertex	Y : string of length 3	Z : point of $\mathbb{R}^3$
$(0, 0, 0)$	000	the origin
$(1, 0, 0)$	100	one step along x
$(0, 1, 0)$	010	one step along y
$(0, 0, 1)$	001	one step along z
$(1, 1, 0)$	110	...
$(1, 0, 1)$	101	...
$(0, 1, 1)$	011	...
$(1, 1, 1)$	111	the opposite corner
path metric	digits that differ	taxicab metric

Three descriptions, three names, one metric space: this is Exercise 21. Two spaces related this way are called *isometric*, and a bijection preserving all distances is an *isometry*. The isometries of a space X form a group $\text{Isom}(X)$, which is the metric analogue of $\text{Aut}(\Gamma)$.

Why?

Why the taxicab metric is the right one on Z . Two corners of the unit cube differ in some set of coordinates, each by exactly 1, so their taxicab distance is the number of coordinates in which they differ. That is the Hamming count again. Had we used the Euclidean metric on Z instead, the opposite corners would be at distance $\sqrt{3}$, and no graph has an irrational distance in its path metric.

Try it

Compute the three pairwise distances between 000, 011 and 111 in all three descriptions and check they agree. Then generalise: the n -dimensional cube graph, the strings of length n , and the unit cube in $\mathbb{R}^n$ with the taxicab metric. What is the largest distance in each?

- 1 are isometric. To see this we put the cube graph in $\mathbb{R}^3$ so that it exactly corresponds to the vertices and edges of the standard unit cube in $\mathbb{R}^3$, with all coordinates equal to 0 or 1. This allows us to label each vertex of the graph by its coordinates in $\mathbb{R}^3$. Then the isomorphism $X \rightarrow Y$ is given by $(x, y, z) \rightarrow xyz$; so, for instance, the vertex $(1, 0, 1)$ gets sent to the string 101. Edges of the cube graph correspond to pairs of vertices that differ in one coordinate—exactly the situation where the corresponding binary strings have distance 1.

The previous discussion suggests a third metric space isometric to both X and Y . Let Z be the vertices of the unit cube in $\mathbb{R}^3$, as above, and endow this set with the taxicab metric in $\mathbb{R}^3$.

- 2 **Exercise 21.** Complete the proof that X , Y , and Z are isometric. Generalize to higher dimensions.

Exercise 22. Let X be $\mathbb{R}$ with the usual metric, and let Y be any line in $\mathbb{R}^n$ with the Euclidean metric on $\mathbb{R}^n$. Show that X and Y are isometric.

- 3 **Isometries of a metric space.** Next, an *isometry* of a metric space X is an isometry $X \rightarrow X$ (as above, isometries are to metric spaces what automorphisms are to groups and graphs). The set of isometries of a metric space forms a group, which we denote by $\text{Isom}(X)$. This is the group of symmetries of the metric space X . Here are some examples:

- 4
1. Rotations about a point in $\mathbb{R}^n$ and translations by some vector in $\mathbb{R}^n$ are both examples of isometries of $\mathbb{R}^n$ with the Euclidean metric.
 2. Reflection across a hyperplane (that is, an $(n-1)$ -dimensional plane) in Euclidean n -space is an isometry. Piggott's Office Hour 13 on Coxeter groups studies groups generated by reflections.
 3. In the taxicab metric on $\mathbb{R}^2$, translations are isometries.
 4. In the hub metric on $\mathbb{R}^2$, rotations about the origin and reflections about lines through the origin are isometries.
 5. Here are two isometries of $C([0, 1])$:

$$\rho_1(f)(x) = -f(x) \quad \text{and} \quad \rho_2(f)(x) = f(1-x).$$

- 5
6. Let's consider now the Hamming distance on the set of binary strings of length n . First, there is an isometry that swaps $0 \leftrightarrow 1$ in each string. Permuting the digits of all strings (by a fixed permutation) defines an isometry too; thus S_n is a subgroup of the group of isometries.

- 6 **Exercise 23.** Check that each of the above functions is an isometry. In each case, try to find other isometries. Or even better: compute the entire isometry group of the given metric space.

Exercise 24. Let X be the subspace of functions $f \in C([0, 1])$ such that $f(0) = f(1)$. Every isometry of $C([0, 1])$ that preserves X is an isometry of X . Can you find additional isometries of X ?

1 Cube, strings, coordinates

An edge of the cube joins two corners differing in exactly one coordinate; two strings at Hamming distance 1 differ in exactly one digit; so the labelling $(x, y, z) \mapsto xyz$ sends edges to pairs at distance 1, and then paths of length k to pairs at distance k . Distances are preserved: an isometry.

2 Exercises 21, 22

(21) For 0/1 vectors, the taxicab distance $\sum |a_i - b_i|$ counts the coordinates that differ, i.e. the Hamming distance; and the same labelling works for the n -cube Q_n . (22) Write the line as $\{p+tu : t \in \mathbb{R}\}$ with $|u| = 1$; then $t \mapsto p + tu$ preserves distances because $|(p+tu) - (p+t'u)| = |t-t'| |u| = |t-t'|$.

3 $\text{Isom}(X)$ is a group

A composition of distance-preserving bijections preserves distances; so does the inverse. Same argument as for $\text{Aut}(\Gamma)$ on page 27. The three “symmetry groups” of the book, S_X , $\text{Aut}(\Gamma)$ and $\text{Isom}(X)$, are all groups of structure-preserving bijections.

4 Examples 1–4

Euclidean: translations, rotations, reflections, and every isometry of $\mathbb{R}^n$ is a composition of these (a theorem, not needed here). Taxicab: translations, and also reflections in the axes and diagonals and rotations by multiples of 90° , but *not* rotation by 45° , which turns the diamond into a square. Hub: only maps fixing the hub can be isometries (the hub is the unique point through which all shortest paths pass), so rotations and reflections about the origin.

5 Examples 5, 6

ρ_1 flips the sign of the function, ρ_2 reverses the interval; both leave the area between two graphs unchanged. For Hamming distance: flipping all digits in a fixed position, or in any fixed set of positions, and permuting positions; together these give $2^n n!$ isometries, which is all of them, and it is also $\text{Aut}(Q_n)$ from V3, as it must be since the metric is the path metric of Q_n .

6 Exercises 23, 24

For 24: functions with $f(0) = f(1)$ are functions on a circle, so “rotate the circle” $f(x) \mapsto f(x+c \bmod 1)$ is an extra isometry of X that does not extend to $C([0, 1])$.

- 1 Group actions on metric spaces.** We said that we should regard $\text{Isom}(X)$ as the group of symmetries of a metric space X . So what does it mean for a group G to act by symmetries on X ? Well, as in the case of graphs, it should mean we have a homomorphism

$$G \rightarrow \text{Isom}(X).$$

If we have such an action $G \rightarrow \text{Isom}(X)$, then each $g \in G$ gives an isometry $X \rightarrow X$. As is usual with group actions, we will write the image of x under this isometry as $g \cdot x$ or gx .

Again unraveling the definitions, a group action of G on the metric space X is a map

$$G \times X \rightarrow X$$

where the image of (g, x) is written $g \cdot x$ and where the following properties are satisfied:

- 2**
- $1 \cdot x = x$ for all $x \in X$,
 - $g \cdot (h \cdot x) = (gh) \cdot x$ for all $g, h \in G$ and $x \in X$, and
 - $d(g \cdot x, g \cdot y) = d(x, y)$ for all $g \in G$ and $x, y \in X$.

The first two items form the definition of a group action of G on X and the third item makes it an action by isometries.

Exercise 25. Verify that our two definitions of a group action by isometries are equivalent.

What are some examples of group actions by isometries?

- 3**
- The dihedral group D_n acts by isometries on a regular n -gon in the plane.
 - The symmetric group S_n acts by isometries on a regular n -simplex by permuting the vertices.¹
- 4**
- Every group acts by isometries on itself with the word metric by left multiplication.
 - $\mathbb{Z}^2$ acts by isometries on $\mathbb{R}^2$ in the Euclidean metric or the taxicab metric by translation.
 - $\mathbb{Z}/n\mathbb{Z}$ acts by isometries on $\mathbb{R}^2$ with the Euclidean metric or the hub metric (via rotation).
 - Every group acts by isometries on every metric space by the *trivial action*: $g \cdot x = x$ for all $g \in G$ and all $x \in X$.
- 5**
- Of course, the last example is not so interesting. It tells us nothing about G or X , as every element of G maps to the identity element of $\text{Isom}(X)$. Recall that the kernel of a homomorphism is the preimage of the identity. In general, we can expect an action of G on X to yield more information if the kernel of the corresponding homomorphism $G \rightarrow \text{Isom}(X)$ is small. In other words, a group action of G on X is most useful when most elements of G do something to X .

6 ¹ A regular n -simplex is the smallest convex shape containing the points in $\mathbb{R}^{n+1}$ that lie on the positive part of a coordinate axis and have distance 1 from the origin; so, for instance, a regular 2-simplex is an equilateral triangle, and a regular 3-simplex is a regular tetrahedron.

1 Action by isometries = homomorphism $G \rightarrow \text{Isom}(X)$

The third time the pattern appears: sets (S_X , page 22), graphs ($\text{Aut}(\Gamma)$, page 28), metric spaces ($\text{Isom}(X)$). The definition is always “a homomorphism into the symmetry group of the object”.

2 Unravelling

The first two bullets are an action on the set X ; the third says each g preserves distances. Exercise 25: the two descriptions match exactly as on page 22 (annotation **3** there).

3 Examples

D_n on the polygon and $\mathbb{Z}^2$ on the plane by translation are the familiar ones. S_n on the regular simplex: the simplex has $n + 1$ vertices all at the same mutual distance, so any permutation of vertices extends to an isometry (V7 draws the simplices of the footnote).

4 A group acts on itself by isometries

Left multiplication is an isometry of the word metric because $d(kg, kh) = d(g, h)$ (page 37, annotation **2**). This is Theorem 2.2 in metric language: G acts on its own Cayley graph, now viewed as a metric space, by distance-preserving symmetries.

5 The kernel of an action

The kernel of $G \rightarrow \text{Isom}(X)$ is the set of elements acting as the identity on X . Trivial action: kernel is all of G . Injective homomorphism: kernel $\{1\}$, an action usually called *faithful* (a word the book does not use). A free action is faithful, since every $g \neq 1$ moves every point; the converse fails (a rotation of the plane is faithful but fixes the centre).

6 Footnote: the regular simplex

Take the $n + 1$ standard basis vectors of $\mathbb{R}^{n+1}$; they are pairwise at distance $\sqrt{2}$. Their convex hull is the regular n -simplex: segment, equilateral triangle, regular tetrahedron, ... (V7).

- 1 **From the geometric to the algebraic.** The next theorem shows the power of applying geometry to the study of groups. In the statement, a *torsion* element of a group is an element of finite order, and a group is *torsion free* if its only torsion element is the identity.

- 2 **THEOREM 2.3.** Suppose that G is a group that has a free action by isometries on $\mathbb{R}^n$ with the Euclidean metric. Then G is torsion free.

This theorem really typifies the theme of this book, as it uses the interplay between the G and the metric space $\mathbb{R}^n$ in order to gain purely algebraic information about G .

- 3 **Proof.** Suppose that $g \in G$ has finite order m and let $\langle g \rangle$ denote the (cyclic) subgroup of G generated by g . We would like to show that $g = 1$. Let v be an element of $\mathbb{R}^n$ and consider the orbit of v under $\langle g \rangle$:

$$\mathcal{O} = \{v, g \cdot v, \dots, g^{m-1} \cdot v\}.$$

- 4 Every finite set in $\mathbb{R}^n$ has a unique *centroid*, that is, there is a unique point w that minimizes the sum of the distances to the points of $\mathcal{O}$:

$$\sum_{i=1}^m d(w, g^i \cdot v).$$

Notice that

$$g \cdot \mathcal{O} = \{g \cdot v, g^2 \cdot v, \dots, g^m \cdot v\} = \mathcal{O}.$$

In other words, the orbit of v under $\langle g \rangle$ is preserved by g (this is true for any group action). It follows that the centroid of $g \cdot \mathcal{O}$ is again w . But since G acts by isometries the centroid of $g \cdot \mathcal{O}$ is $g \cdot w$, and so this means $g \cdot w = w$. Since we assumed that the action of G is free, it must be that $g = 1$, as desired. $\square$

- 5 **Exercise 26.** Prove that any finite set of points in $\mathbb{R}^n$ has a unique centroid. *Hint: Convex functions have unique minima and the sum of convex functions is convex.*

2.3 GEOMETRIC GROUP THEORY: GROUPS AND THEIR SPACES

- 6 In the first two office hours we have discussed two types of objects: groups and spaces. The overarching question in geometric group theory is:

Which groups act by isometries on which spaces? If a group does act on a space, what does it say about the group? What does it say about the space? If we think of a group as a metric space, what does it look like?

1 Torsion

An element has *finite order* m if $g^m = 1$ with $m \geq 1$ smallest; such elements are torsion. $\mathbb{Z}/n\mathbb{Z}$, D_n , S_n consist of (or contain) torsion; $\mathbb{Z}^n$ and F_n are torsion-free (in F_n , powers of a reduced word never cancel completely).

2 Theorem 2.3

A group acting *freely* by isometries on Euclidean space cannot contain elements of finite order. So D_n , or any finite group, cannot act freely on $\mathbb{R}^n$, even though they act by rotations and reflections; those actions have fixed points. $\mathbb{Z}^n$ acts freely by translations, consistent with the theorem.

Big picture

This is the first theorem of the “geometric $\Rightarrow$ algebraic” type: the hypothesis is about a space and an action, the conclusion is pure algebra. The book is full of these; the last page lists two more.

3 The proof, step by step

(i) Suppose g has finite order m . (ii) The orbit $\mathcal{O}$ of any point v under $\langle g \rangle$ is a finite set. (iii) $\mathcal{O}$ has a unique “centre” w . (iv) g permutes $\mathcal{O}$, so $g \cdot \mathcal{O} = \mathcal{O}$. (v) An isometry sends the centre of $\mathcal{O}$ to the centre of $g \cdot \mathcal{O}$; hence $g \cdot w$ is the centre of $\mathcal{O}$, i.e. $g \cdot w = w$. (vi) Freeness forces $g = 1$. V7 draws the picture: for a rotation of order 5 the orbit is a pentagon and its centre is fixed.

4 Which “centroid”?

Watch out

As printed, the centroid is the point minimising the *sum of distances*. That minimiser is not always unique: for two points, every point of the segment between them minimises $d(w, p_1) + d(w, p_2)$. The proof only needs *some* rule that assigns to each finite set a single point and commutes with isometries. The ordinary average $\frac{1}{m} \sum g^i \cdot v$ works: it minimises the sum of *squared* distances, whose minimiser is unique (the function is strictly convex), and Euclidean isometries send averages to averages because they are affine maps. With that reading Exercise 26 is exactly this uniqueness.

5 Exercise 26

Show $w \mapsto \sum_i |w - p_i|^2$ is strictly convex and expand to see it is minimised at the average. (Convex functions need not have unique minima; strictly convex ones do.)

6 The overarching question

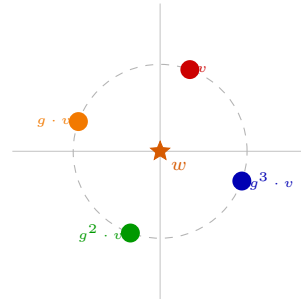
Four questions in one. Read them again after each later office hour; every chapter answers a piece of one of them.

Interlude: K. Page 40: geometry forcing algebra, and the centroid trick behind it

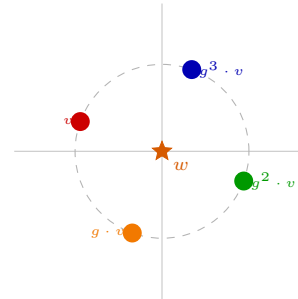
Theorem 2.3 is the first result in the book where a purely geometric hypothesis yields a purely algebraic conclusion: *if G acts freely by isometries on $\mathbb{R}^n$ with the Euclidean metric, then G is torsion free*, meaning no element except the identity has finite order. The proof is two lines long once the right object is in view, and the object is the centroid of a finite set.

The argument, before and after

Suppose g has finite order m . Pick any point v and look at its orbit under the powers of g , a set of at most m points. Mark its centroid, the average of the points.



the orbit of v under $\langle g \rangle$, here of order 4, and its centroid w



after applying g : every point of the orbit has moved to the next one, but the centroid has not moved at all

Applying g permutes the orbit, because g carries $g^k \cdot v$ to $g^{k+1} \cdot v$ and the last point back to the first. A permutation of a finite set does not change its average, so g fixes the centroid w . But the action is free, and the only element allowed to fix anything is the identity. Hence $g = 1$, and the only element of finite order is the identity.

Why?

Why an isometry of $\mathbb{R}^n$ moves centroids to centroids. Every isometry of Euclidean space is an affine map: a rotation or reflection followed by a translation. Affine maps commute with averaging, so the image of the average is the average of the images. This is the one place the hypothesis “ $\mathbb{R}^n$ with the Euclidean metric” is used, and it is why the theorem is stated for that space rather than for metric spaces in general.

Watch out

Both hypotheses are doing work. Drop *free*: the rotation by a quarter turn about the origin generates a group of order 4 acting by isometries on $\mathbb{R}^2$, with plenty of torsion. It fixes the origin, so it is not free. Change the *space*: the same rotation group acts freely on the circle by isometries, and it is not torsion free. So the conclusion is not about free actions in general but about free actions on Euclidean space, where centroids exist.

Big picture

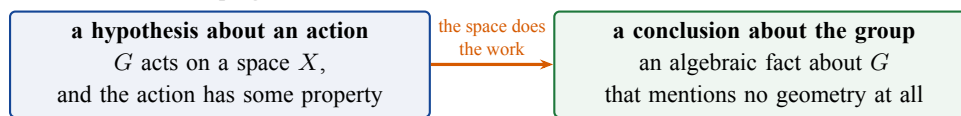
This is the shape of every theorem the book is heading towards. From “ G acts in a prescribed way on a space of a prescribed kind” one deduces an algebraic fact about G that mentions no geometry at all. Office Hour 3 does it with trees: a group acting freely on a tree is a free group. Office Hour 5 has more. The reason such theorems are available is Theorem 2.2: every group has a space to act on, so the hypotheses are never vacuous.

Try it

Where exactly does the argument break if g has infinite order? (Look at what the orbit becomes, and whether it has a centroid.) Then use the theorem to conclude that $\mathbb{Z}/n\mathbb{Z}$ has no free action by isometries on any $\mathbb{R}^n$, and check that this is consistent with the rotation example above.

Interlude: M. Geometry forcing algebra: one pattern, five instances

Every theorem below has the same shape. You are told how a group acts on some space, you are told one property of that action, and out comes a fact about the group with no geometry in it at all. This is the reason the office hour spends so long on definitions that look like bookkeeping.



1. Cayley's theorem (page 23)

Action: G on itself by left multiplication. *Property:* free.

Conclusion: G is isomorphic to a subgroup of a symmetric group.

Why: freeness makes the homomorphism $G \rightarrow S_G$ injective, so G is a group of permutations whether or not it was defined as one.

2. Lagrange's theorem

Action: G on the cosets of a subgroup H , by left multiplication. *Property:* transitive, with stabiliser H .

Conclusion: $|H|$ divides $|G|$.

Why: orbit–stabiliser reads $|G| = |\text{cosets}| \cdot |H|$. A counting theorem about subgroups falls straight out of a counting theorem about actions.

3. Groups of order p^2 are abelian

Action: G on itself by conjugation. *Property:* the orbits are the conjugacy classes, and each has size dividing $|G|$.

Conclusion: a group whose order is a power of a prime has a non-trivial centre; and if the order is p^2 the group is abelian.

Why: adding up the orbit sizes gives $|G|$. Every class of size bigger than one has size divisible by p , and so does $|G|$, so the classes of size one cannot number just one. Those classes are exactly the central elements.

Big picture

Note what happened in the third example. The statement “every group of order p^2 is abelian” contains no geometry, no space and no distance. It is proved by making the group act on itself and counting orbits. This is the cheapest possible illustration of the method, and it needs nothing from this office hour beyond the definition of an action.

4. Free isometric actions on $\mathbb{R}^n$ (page 40)

Action: G on $\mathbb{R}^n$ with the Euclidean metric. *Property:* free, and by isometries.

Conclusion: G is torsion free, meaning no element except the identity has finite order.

Why: a finite orbit would have a centroid, and an isometry of Euclidean space moves centroids to centroids, so the element would fix a point. Interlude K draws it.

Used for: $\mathbb{Z}/n\mathbb{Z}$ can never act freely by isometries on any $\mathbb{R}^n$. Neither can any group containing an element of finite order, which rules out most finite groups at a glance.

5. Free actions on trees (Office Hour 3)

Action: G on a tree. *Property:* free.

Conclusion: G is a free group.

Why: freeness prevents any nontrivial element from fixing a vertex, and on a tree that forces each element to translate along an axis; the combinatorics then produces a free generating set.

Used for: the showcase corollary. Every subgroup of a free group is free. The proof is two lines once the theorem is available: F_2 acts freely on its Cayley tree, a subgroup $H \leq F_2$ acts freely on that same tree by restriction, so H is free. The statement is pure algebra and the argument is pure geometry.

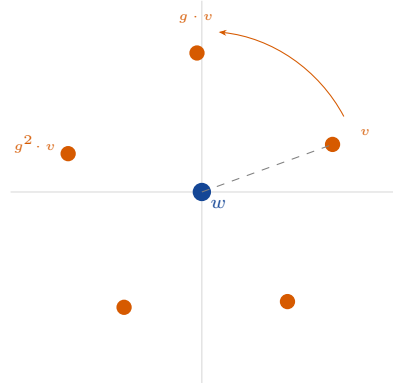
Looking ahead

Office Hour 7 carries the pattern further with a theorem that runs in both directions: if a group acts on a space in a sufficiently controlled way, then the group and the space have the same large-scale shape. At that point the word metric of Interlude J stops being a curiosity and becomes the object of study.

Try it

Use instance 4 to decide which of these can act freely by isometries on the plane: $\mathbb{Z}$, $\mathbb{Z}^2$, $\mathbb{Z}/6\mathbb{Z}$, D_4 , F_2 . For the ones that can, describe the action; for the ones that cannot, name an element of finite order. Then use instance 5 to explain why the subgroup of F_2 generated by a^2 and bab^{-1} is free, without computing anything.

Interlude: V7. Theorem 2.3 in a picture, distances in a small group, and the map of the chapter



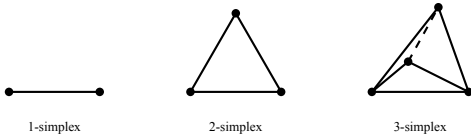
g of order 5, acting by rotation. Its orbit O is a regular pentagon; the average w of O is fixed by g . A *free* action forbids fixed points, so no such g exists.

Word metric on $D_3 = \langle s, t \rangle$ (Exercise 20).

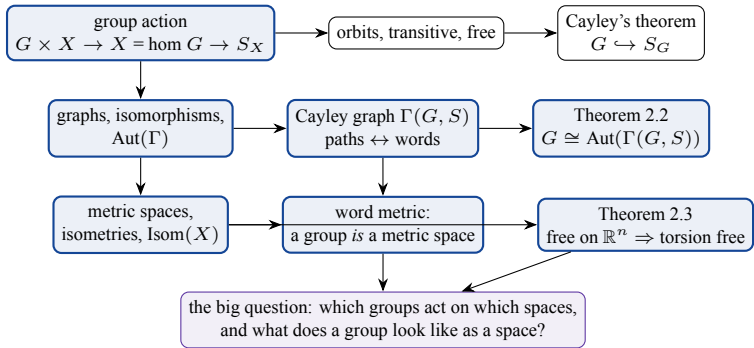
d	1	s	t	st	ts	sts
1	0	1	1	2	2	3
s	1	0	2	1	3	2
t	1	2	0	3	1	2
st	2	1	3	0	2	1
ts	2	3	1	2	0	1
sts	3	2	2	1	1	0

st and ts are the two rotations, $sts = tst$ is the third reflection. Read distances off the hexagon of page 29.

Regular simplices (page 39, footnote).



S_{n+1} permutes the $n + 1$ vertices; every permutation is an isometry.



Looking ahead
Every later office hour sits somewhere on this map: Office Hours 3–6 on the “Cayley graph of F_2 is a tree” corner, 7–12 on the “group as metric space” corner, and 13–18 on specific groups and their spaces.

We already saw a first example of this in Theorem 2.3. Another theorem in this vein will be discussed in Office Hour 3:

1 *If a group acts freely on a tree, then the group is a free group.*

Like Theorem 2.3, this theorem typifies geometric group theory: from just the information that our group acts on a certain kind of space in a certain way, we deduce concrete algebraic information about the group. There are similar theorems in Office Hour 5: if a group acts in a prescribed way on certain sets, graphs, or metric spaces, we can deduce that the group is free.

Here is a theorem from Office Hour 7, regarding the geometry of a group thought of as a metric space via the word metric.

2 *If a group is quasi-isometric to $\mathbb{Z}$, then it has a subgroup of finite index that is isomorphic to $\mathbb{Z}$.*

We'll define quasi-isometric in Office Hour 7, but for now you can read it as "roughly isometric." The point of the theorem is that rough geometric information about a group can tell us fine algebraic information about the group. We'd like to emphasize that we could not have even formulated this theorem without our new perspective on groups as metric spaces.

3 Geometric group theory is full of beautiful examples of groups and spaces and interactions between the two. But don't take our word for it—get going!

1 Free actions on trees give free groups

The cousin of Theorem 2.3 for trees instead of $\mathbb{R}^n$: proved in Office Hour 3. Example already in hand: F_2 acts freely on its Cayley graph T_4 , and it is free. The theorem says that this is no accident: *any* group that acts freely on *any* tree is a free group. Applied to subgroups of F_2 (they act freely on T_4 as well), it gives the famous fact that every subgroup of a free group is free.

2 Quasi-isometric to $\mathbb{Z}$

"Quasi-isometric" will mean "looks the same from far away" (Office Hour 7). "Finite index" means the subgroup has only finitely many cosets. Example: the symmetry group of the undecorated integer line, translations and reflections (the infinite dihedral group D_∞), looks like a line from far away and contains the translations $\mathbb{Z}$ as a subgroup of index 2. The theorem says this always happens: a group shaped like a line is, up to finite index, the group $\mathbb{Z}$.

Big picture

Both theorems have the same form: a *geometric* hypothesis (acts freely on a tree; looks like a line) and an *algebraic* conclusion (is free; contains $\mathbb{Z}$ with finite index). Neither could be stated before groups became spaces on page 37.

3 Get going

What you now have: every group is the symmetry group of its Cayley graph (Theorem 2.2); every group with a generating set is a metric space (page 37); actions on spaces constrain groups (Theorem 2.3). The map on V7 shows how the pieces connect and where the later office hours attach.

Try it

Before moving on, draw the Cayley graph of your favourite group from Office Hour 1 with your favourite generating set, mark the word-length balls of radius 1 and 2, and find the automorphism Φ_g for one nontrivial g . That is the whole chapter in one exercise.

Closing page: all twenty-six exercises at a glance, and a glossary

- 1. (p. 22) Actions = homomorphisms $G \rightarrow S_X$: $\sigma_g(x) = g \cdot x$; axiom 2 gives $\sigma_{gh} = \sigma_g \sigma_h$, axiom 1 gives $\sigma_1 = \text{id}$ so each σ_g is a bijection.
- 2. (p. 22) More actions: $\mathbb{R}^\times$ on $\mathbb{R}$ by scaling; $\text{GL}(n, \mathbb{R})$ on lines through 0; $\mathbb{Z}$ on $\mathbb{Z}$ by $m \cdot n = m + n$; D_n on the edges of the n -gon.
- 3. (p. 22) Orbits: $\{n, -n\}$ for ± 1 on $\mathbb{Z}$; circles for rotations; one orbit for S_n ; $\{x\}$ for the trivial action.
- 4. (p. 23) $g \cdot (h \cdot x) = g(hxh^{-1})g^{-1} = (gh)x(gh)^{-1}$; left multiplication is associativity.
- 5. (p. 23) Conjugacy classes: cycle type in S_n ; $\{r^k, r^{-k}\}$ and one or two classes of reflections in D_n ; cyclic rotations of cyclically reduced words in F_n ; singletons in $\mathbb{Z}^n$.
- 6. (p. 23) Formal graph: list V, E and the endpoint function as a table (V-page 23 annotation).
- 7. (p. 26) Definitions on V2.
- 8. (p. 26) Cube graph isomorphism via binary labels (V3).
- 9. (p. 26) $K_{m,n} \cong K_{n,m}$: swap the colour classes.
- 10. (p. 26) Petersen: label by 2-subsets so that adjacency is disjointness; check invariants first.
- 11. (p. 27) $\text{Aut}(\text{cube graph})$: order 48, rotations and reflections of the cube; $S_4 \times \mathbb{Z}/2\mathbb{Z}$.
- 12. (p. 28) Yes: $|\text{Aut}(\text{Petersen})| = 120 = |S_5|$.
- 13. (p. 28) Slide T_3 along a bi-infinite path.
- 14. (p. 28) Isomorphic components; friends with identical other friends.
- 15. (p. 30) Drawn on V4 (three of the four); S_4 is the permutohedron.
- 16. (p. 30) The labels spell $g^{-1}h$.
- 17. (p. 36) Hub metric: case analysis (page 35 annotation); path metric: concatenate paths.
- 18. (p. 36) Circle, diamond, square; hub: circle about 0 but two points about any other centre (V6).
- 19. (p. 36) Unique: Euclidean, hub. Not unique: taxicab, sup, most graphs.
- 20. (p. 37) D_3 table on V7.
- 21. (p. 38) Taxicab on $\{0, 1\}^n$ counts differing coordinates = Hamming = path metric on Q_n .
- 22. (p. 38) $t \mapsto p + tu$ with $|u| = 1$.
- 23. (p. 38) Full isometry groups: $\mathbb{R}^n$: all compositions of translations, rotations, reflections; Hamming: $2^n n!$; hub: $O(2)$ fixing the origin.
- 24. (p. 38) Rotations $f(x) \mapsto f(x + c \bmod 1)$.
- 25. (p. 39) Same translation as Exercise 1 with S_X replaced by $\text{Isom}(X)$.
- 26. (p. 40) Use the sum of squared distances; strictly convex, minimised at the average.

Glossary (page of first appearance)

action (22)	$G \times X \rightarrow X$ with $1 \cdot x = x, g \cdot (h \cdot x) = (gh) \cdot x$
S_X (22)	all bijections $X \rightarrow X$
orbit (22)	$G \cdot x = \{g \cdot x\}$
transitive / free (22)	one orbit / no fixed points for $g \neq 1$
conjugacy class (23)	orbit of $g \cdot h = ghg^{-1}$
Cayley's theorem (23)	G embeds in S_G
graph, endpoint function (23)	$V, E, E \rightarrow$ unordered pairs of V
valence (23)	number of half-edges at a vertex
$K_n, K_{m,n}$ (24)	complete, complete bipartite
Petersen graph (25)	2-subsets of $\{1..5\}$, edges = disjoint
tree, T_3 (25)	connected, no cycles; regular 3-valent
directed / labelled (26)	ordered endpoints / labels on V or E
graph isomorphism (26)	compatible bijections on V and E
$\text{Aut}(\Gamma)$ (27)	automorphisms = symmetries of Γ
Cayley graph $\Gamma(G, S)$ (29)	vertices G , edges $g \rightarrow gs$ labelled s
word length $ g _S$ (30)	shortest word for g ; path distance from 1
Φ_g (30)	left multiplication, an automorphism of $\Gamma(G, S)$
Theorem 2.2 (30)	$G \cong \text{Aut}(\Gamma(G, S))$
T_4 (32)	Cayley graph of F_2
metric space (34)	set with d : positive definite, symmetric, triangle inequality
Euclidean, taxicab, hub, sup (34–35)	four metrics on $\mathbb{R}^2$
L^1 metric (35)	$\int f - g $ on $C([0, 1])$
Hamming distance (36)	number of differing digits
path metric (36)	fewest edges between vertices
geometric realisation (36)	graph made into a space, edges = $[0, 1]$
word metric (37)	$d(g, h) = g^{-1}h _S$
isometry, $\text{Isom}(X)$ (37–38)	distance-preserving bijection; their group
action by isometries (39)	homomorphism $G \rightarrow \text{Isom}(X)$
torsion, torsion-free (40)	finite-order elements; only 1 has finite order
Theorem 2.3 (40)	free action on $\mathbb{R}^n \Rightarrow$ torsion free
quasi-isometric (41)	roughly isometric; Office Hour 7
finite index (41)	finitely many cosets